

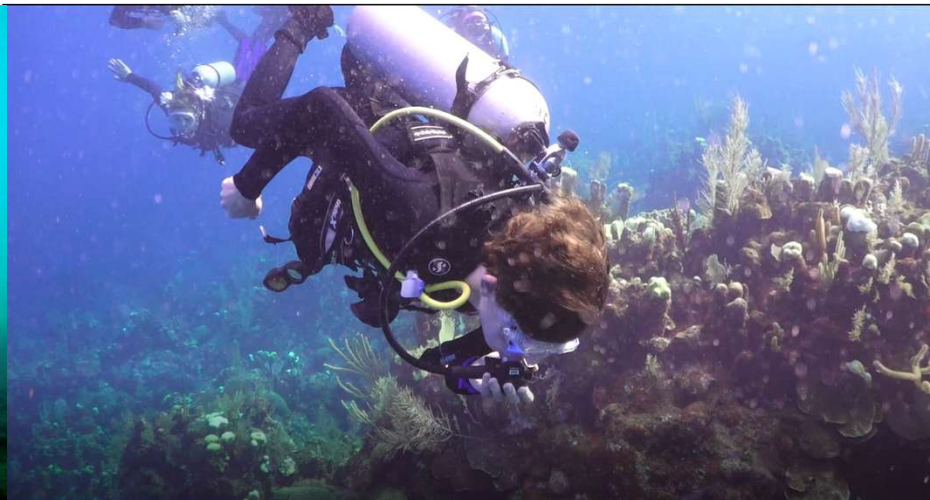
CALPHAD Training

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About Me

- Born & Raised Knoxville, TN
- Eagle Scout
- B.S. In Materials Science and Engineering at The University of Tennessee Knoxville
- I enjoy hiking, camping, backpacking, rock climbing, and European Football (Bayern Munich)
- Scuba certified





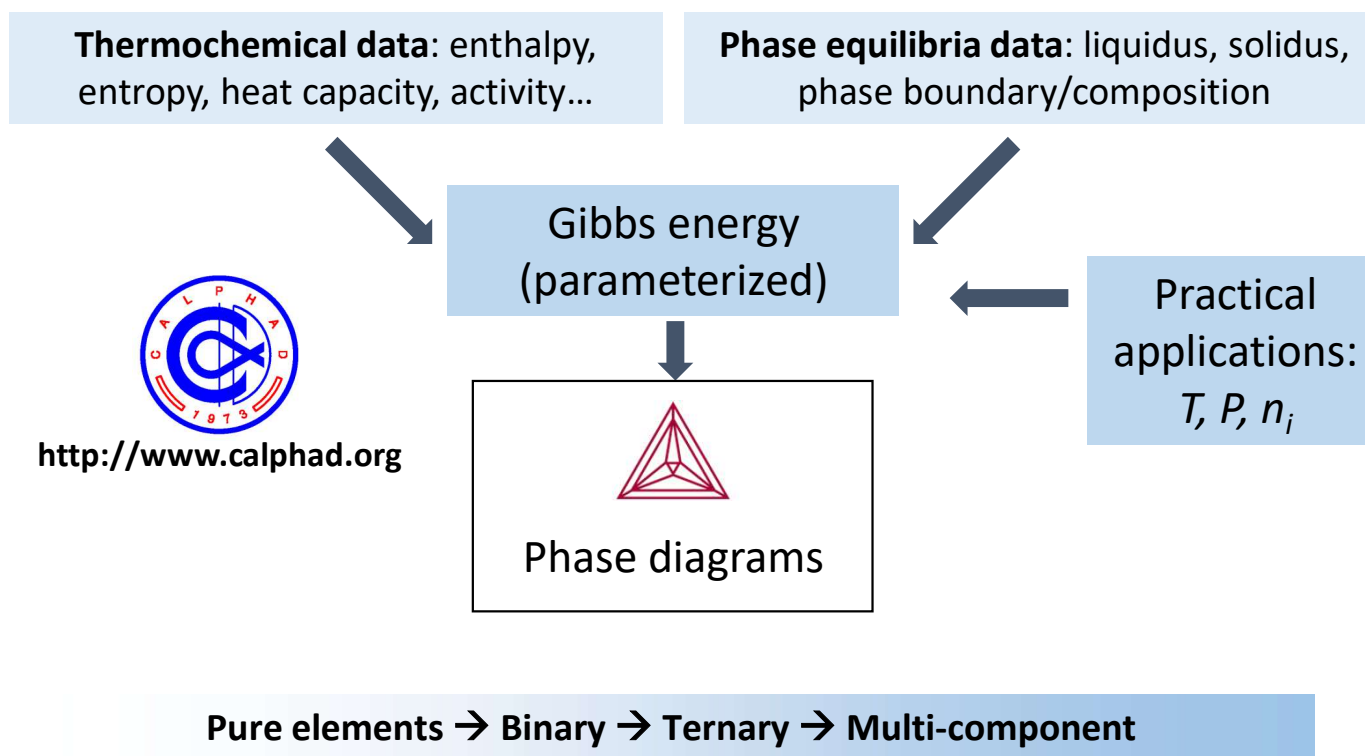
Outline



- CALPHAD
- Simulations of Choice
- Thermodynamic databases
- Pycalphad
- Thermocalc(self-learn)
- AMMap(Future)



CALculations of PHase Diagrams (CALPHAD)



<http://www.calphad.org>

<http://www.calphad.org>

Crash Course in CALPHAD

```
PHASE B2 % 3 .5 .5 3 !  
CONSTITUENT B2 :FE,NI,TI,V : FE,NI,TI,V : VA : !
```

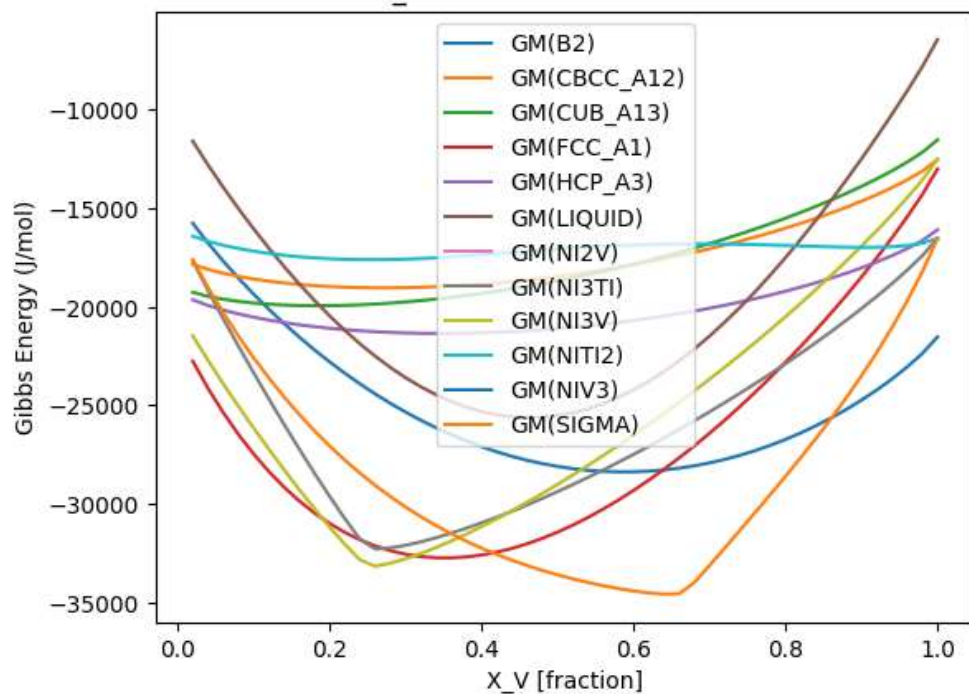
```
PARAMETER G(B2,FE:TI,V:VA;0) 298.15 +VA29#+VA30#*T; 6000 N REF0 !
```

- Phase “B2”
- Model: 3 sublattices, [.5,.5,3]
- What can occupy each sublattice
- One specific component given as a parameter and it's function

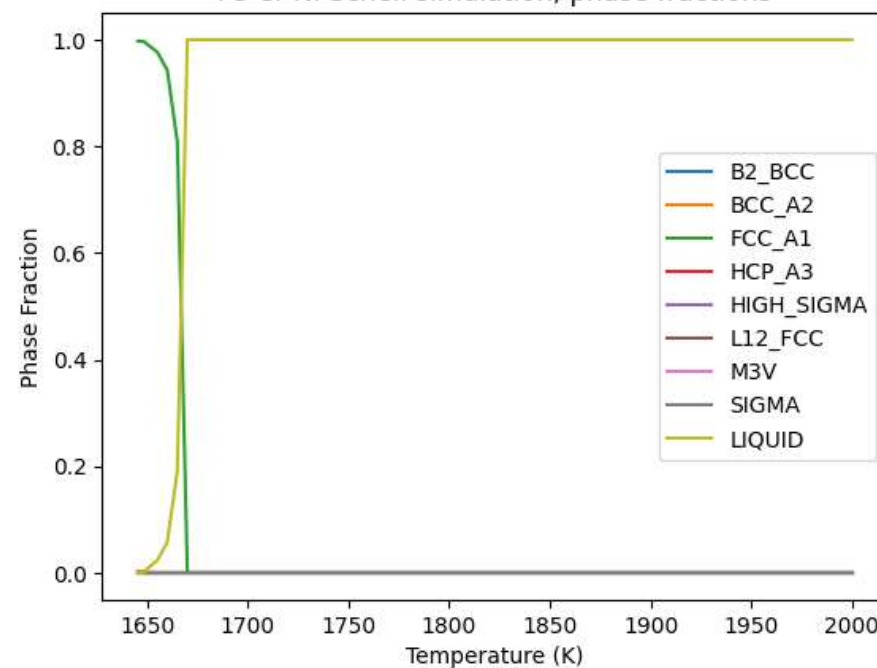
$${}^0G^\varphi(T) - H^{SER} = a + bT + cT\ln(T) + dT^2 + eT^{-1} + fT^3 + gT^7 + hT^{-9}$$

Free energy curve

Ni-V_zou2018 Phases at T=600 K



Fe-Cr-Ni Scheil simulation, phase fractions





Equilibrium Calculations

- Finds the global Gibbs-energy minimum at a fixed temperature and overall composition — the state the alloy would reach after infinite time.
- Assumptions
 - Complete diffusion in every phase, liquid and solid
 - No kinetic barriers: stable phases always form, metastable phases never persist
 - Composition is uniform; in the case study, the nominal composition of each layer



Captures

- Long-time phase stability vs. T
- Phase amounts and compositions
- Fast; needs only a thermodynamic database



Misses

- Microsegregation left by solidification
- Metastable phases (e.g., γ'' in IN625)
- Time: no sense of how fast phases form

Scheil

- Solidification simulation, stepping temperature down until fully solidified, finding the solid-liquid equilibrium at each T
- Assumptions:
 - The liquid is perfectly mixed
 - No solid diffusion
 - solid-liquid interface is at local equilibrium
- Results can be used for hot cracking criteria



Captures

- Microsegregation
- Solidification sequence and freezing range
- Fast processes microstructure

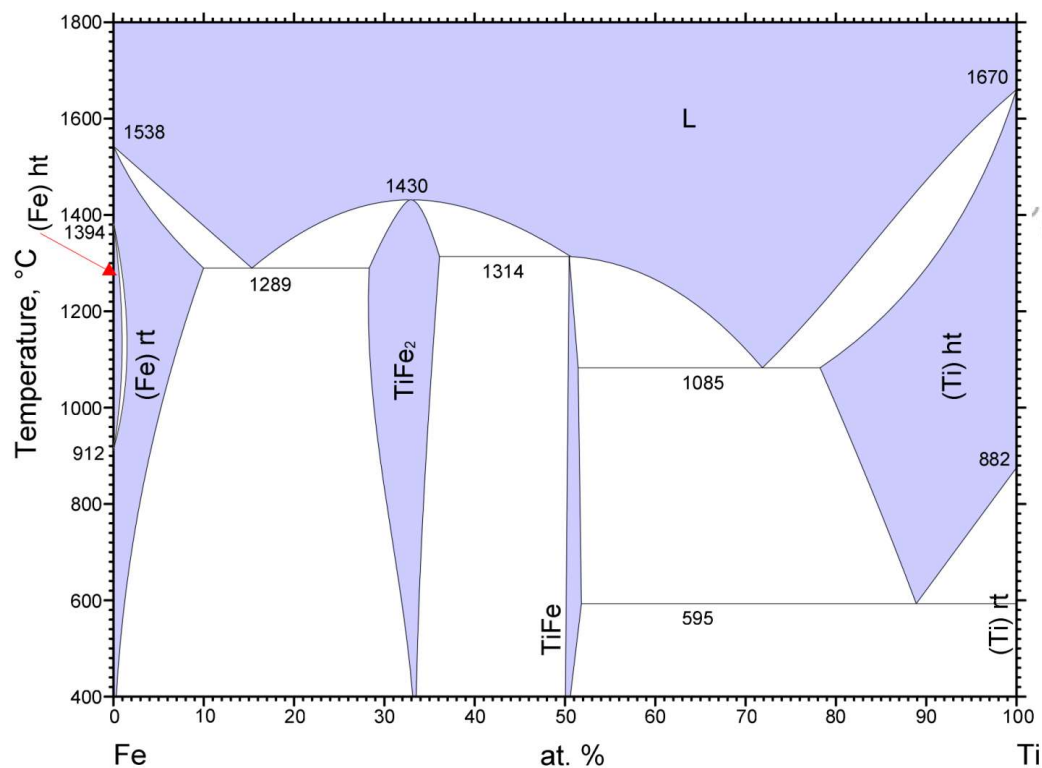


Misses

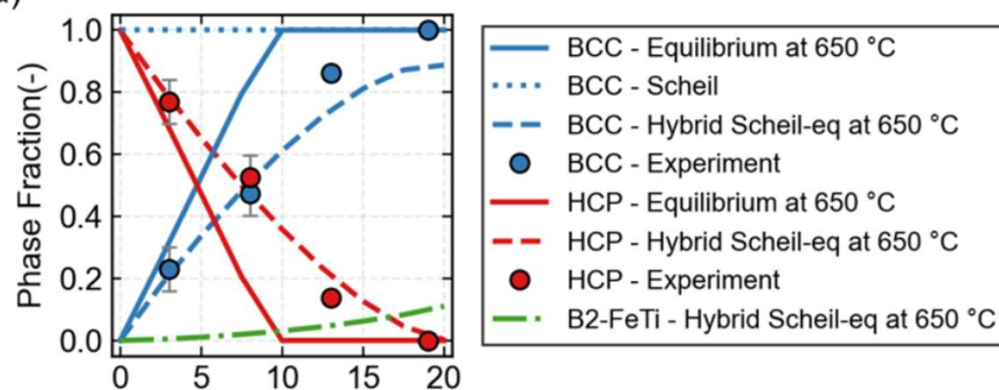
- Solid diffusion
- Solid-state transformation
- Cooling rate effects

Hybrid-Scheil

a Scheil solidification path
considering solidification rather



(a)

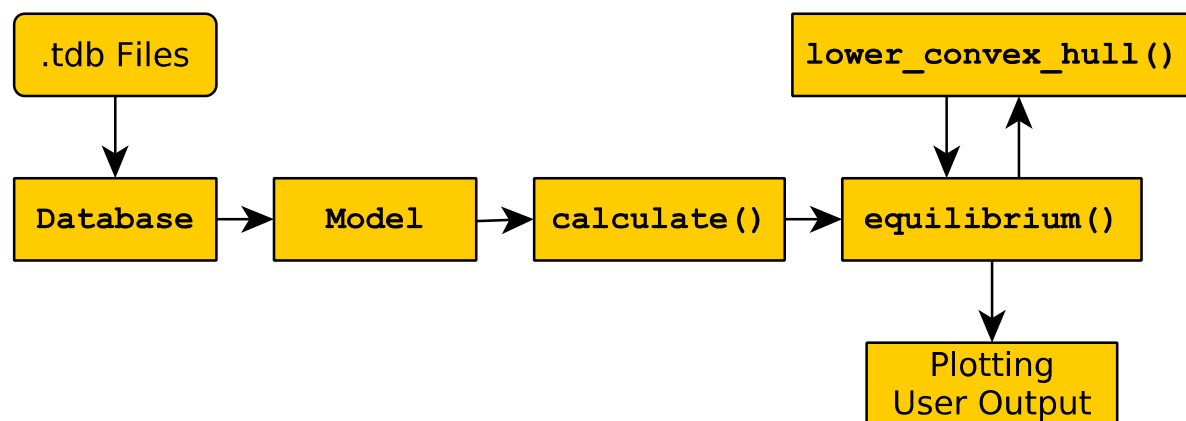




TDB files in the literature

- A. van de Walle, C. Nataraj, Z.K. Liu , “The Thermodynamic Database Database ”, Calphad 61 (2018) 173-178. <https://doi.org/10.1016/j.calphad.2018.04.003>, <https://avdwgroup.engin.brown.edu/>
- NIMS: MatNavi, <https://mits.nims.go.jp/en/>, <https://cpddb.nims.go.jp/en>
- NIST: <https://materialsdata.nist.gov/handle/11256/8>
- MatCalc: <https://www.matcalc.at/index.php/databases/open-databases>

PyCalphad.org

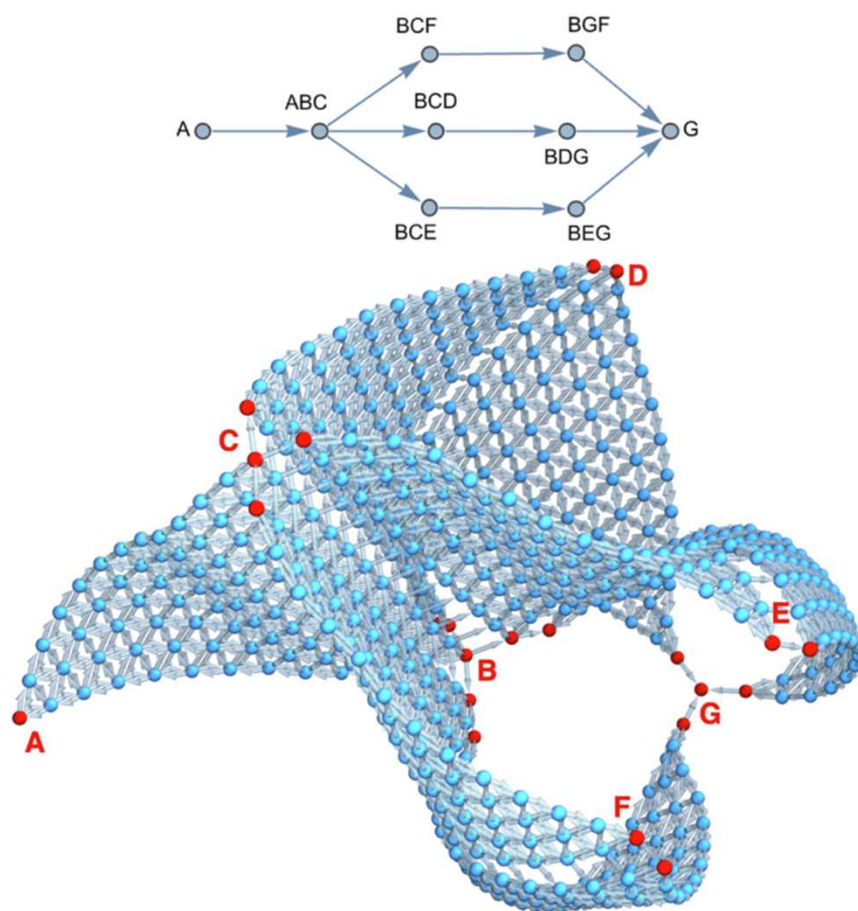


J. Open Res. Softw. 5 (2017) 1

- advanced sampling algorithms
- SymPy: Symbolic computing in python, Custom models
- `calculate()`: Calculate properties with independent ξ
- `equilibrium()`: Free energy minimization
- runner-up (2nd place) NASA Software of the Year 2019, <https://icb.nasa.gov>

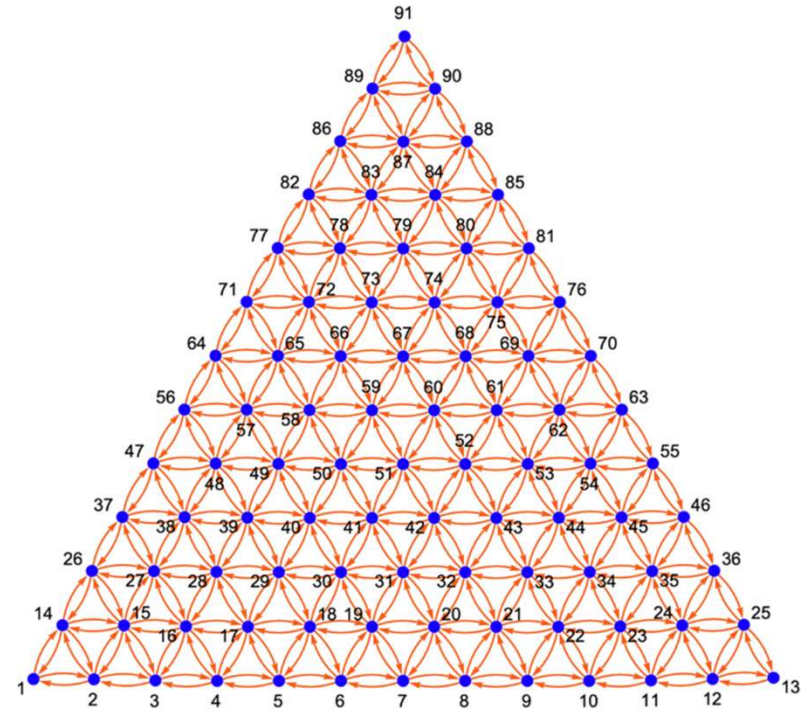
AMMap

- Tool for easy design of complex space for AM CCMs like FGMs
- Highly efficient, ordered space generation with nimplex
- Stacking applications of tools for more complete design considerations
- Framework for implementation of new properties



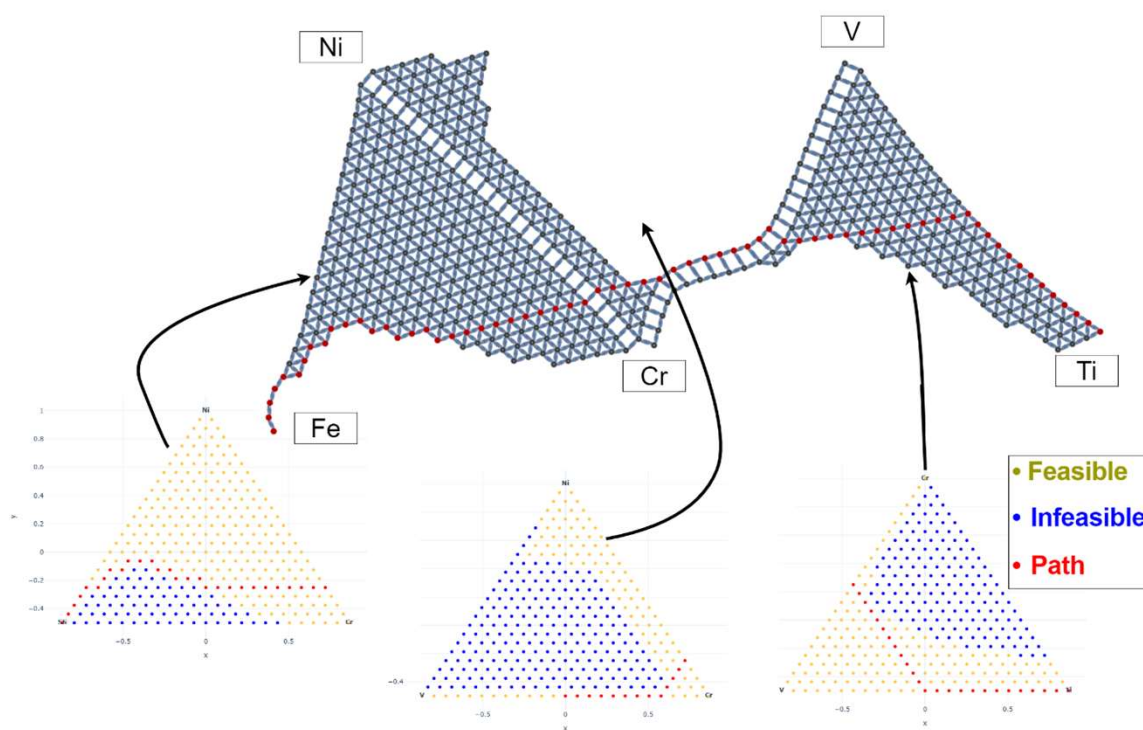
nimplex

- NIM library for simPLEX spaces
- Automatic generation of uniform space and all connectivities of points
- Expandable to n-dimensional space
- Ability to join spaces through shared edges, generating singular spaces from different systems



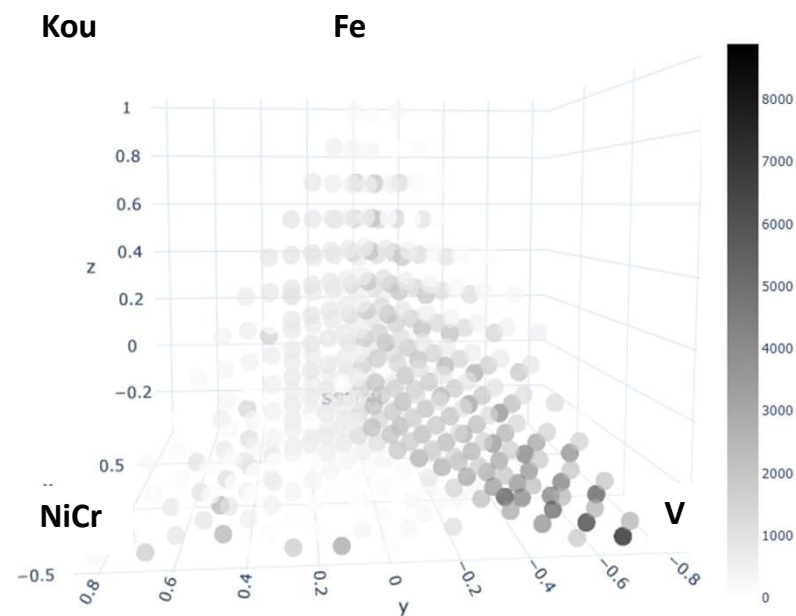
Thermodynamic Simulation

- Equilibrium simulation from CALPHAD tool pycalphad
- Scheil-Gulliver for predicting formation of solid phases based on changing liquid composition
- Infeasibility gliding to reduce calculations performed

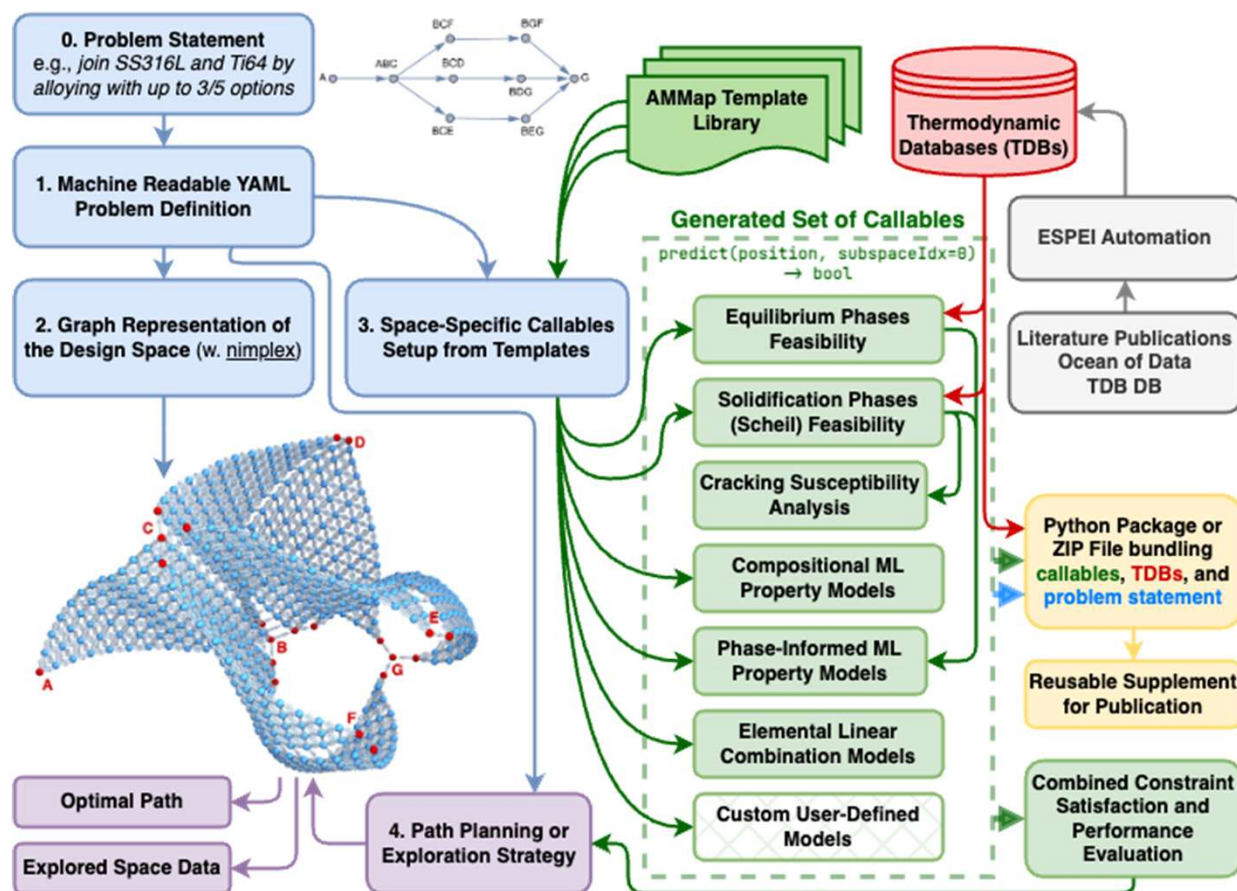


Modular Callables and ML

- Designed for expansion with added properties
- For example, the implemented hot-cracking criteria
- Through use of templates, callables can be made customized to design spaces
- Simplex space is highly compatible with already existing ML tools or models

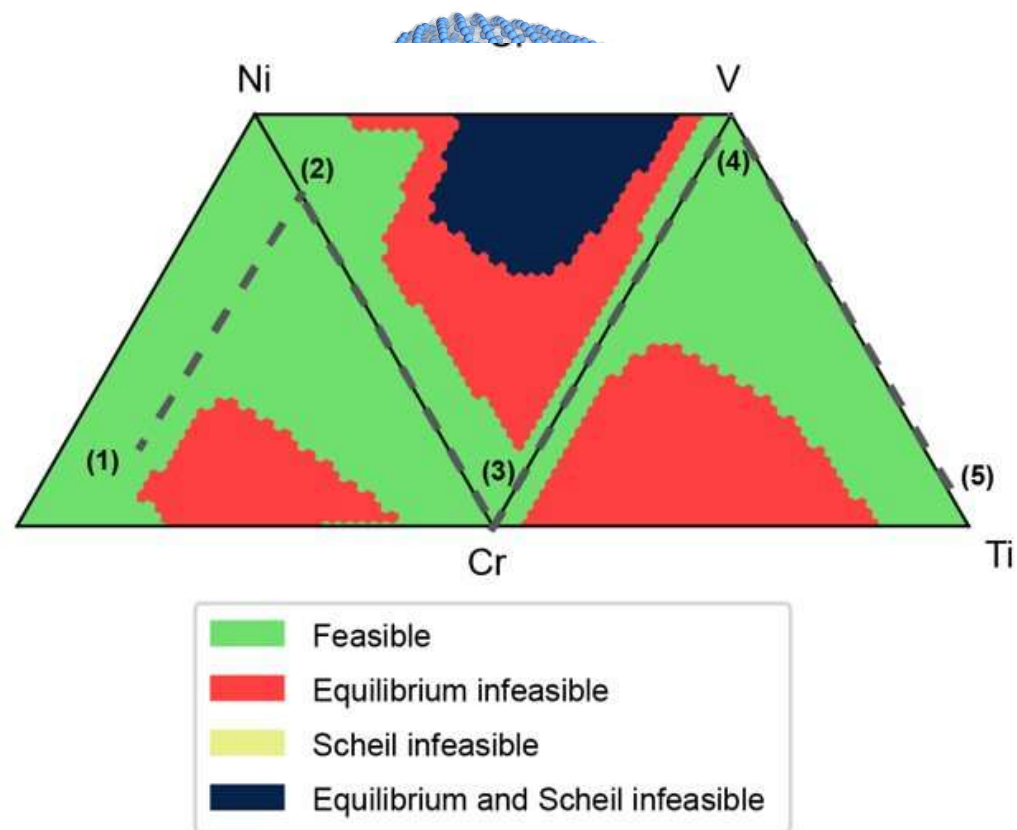


Structure



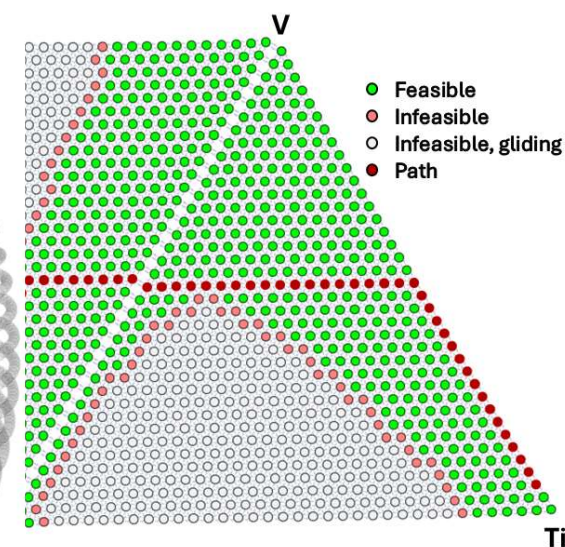
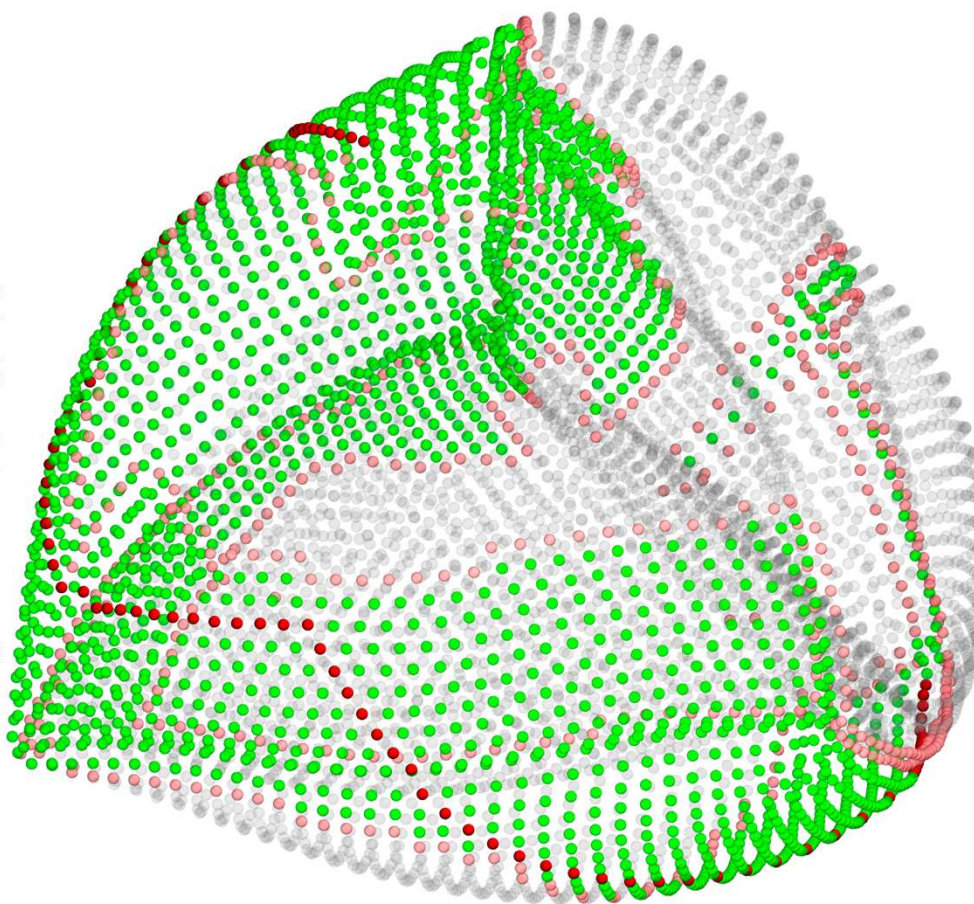
Demonstration

- SS316-Ti64 desirable FGM system previously studied by group
- 10 distinct ternary spaces over 9 thermodynamic databases
- Manual path determination from SS316- $\rightarrow$ Ni-20Cr- $\rightarrow$ Cr- $\rightarrow$ V- $\rightarrow$ Ti64



Demonstration

- Shortest path for simple python algorithm avoiding both Scheil equilibrium infeasibility
- Infeasibility Gliding
 - Avoided Avoided (63.14%) equilibrium simulations
 - Avoided 3729/7 Scheil simulations





Github/codespace examples

- Zhening TC:
[zvy5212/FGM-Code Thermo calc-based: Equilibrium, Scheil, hybrid Scheil-Eq, hot cracking, etc.](#)
- [PhasesResearchLab/AMMap: Additive Manufacturing Mapping of Compositional Spaces with Thermodynamic, Analytical, and Artificial Intelligence Models](#)

Thermo-Calc Modules



Thermo-Calc

Thermodynamic and phase diagram calculations for multicomponent systems.



Precipitation module (TC-PRISMA)

Simulation of 3D multiparticle diffusive precipitation kinetics.



Diffusion module (DICTRA)

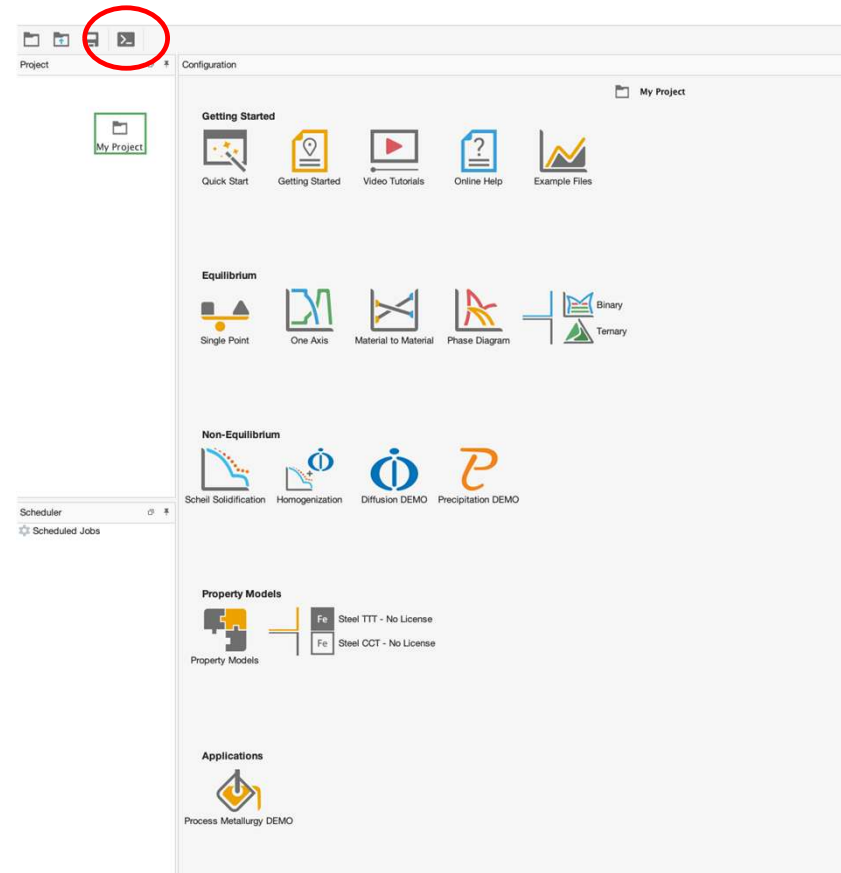
Simulation of Diffusion Controlled phase TRAnsformations.

Outline:

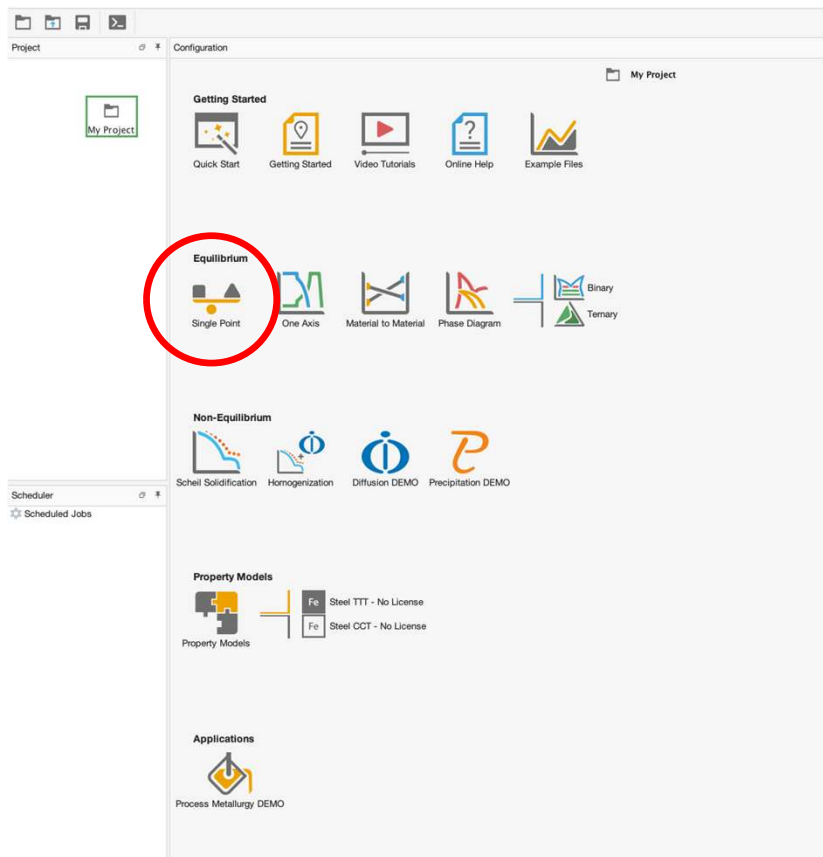
- **One component**
 - **Al:** point calculations
 - **Al:** enthalpy
 - **Ni:** P-T diagram
- **Binary system: Fe-Cr**
- **Ternary system: Fe-Cr-C**
 - Isothermal section: Cr-Fe-C
 - Liquidus projection: Cr-Fe-Ni

Opening graphical mode in Thermo-calc

- Search for Thermo-calc and select “Thermo-Calc” from the menu
- Next select graphical mode in the upper left-hand corner. You screen should look like the one at right:

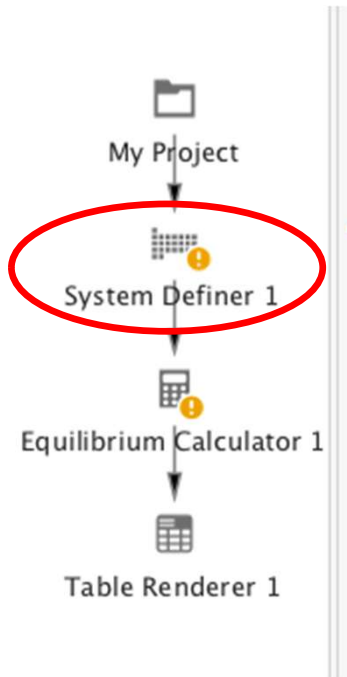


Point Calculation:



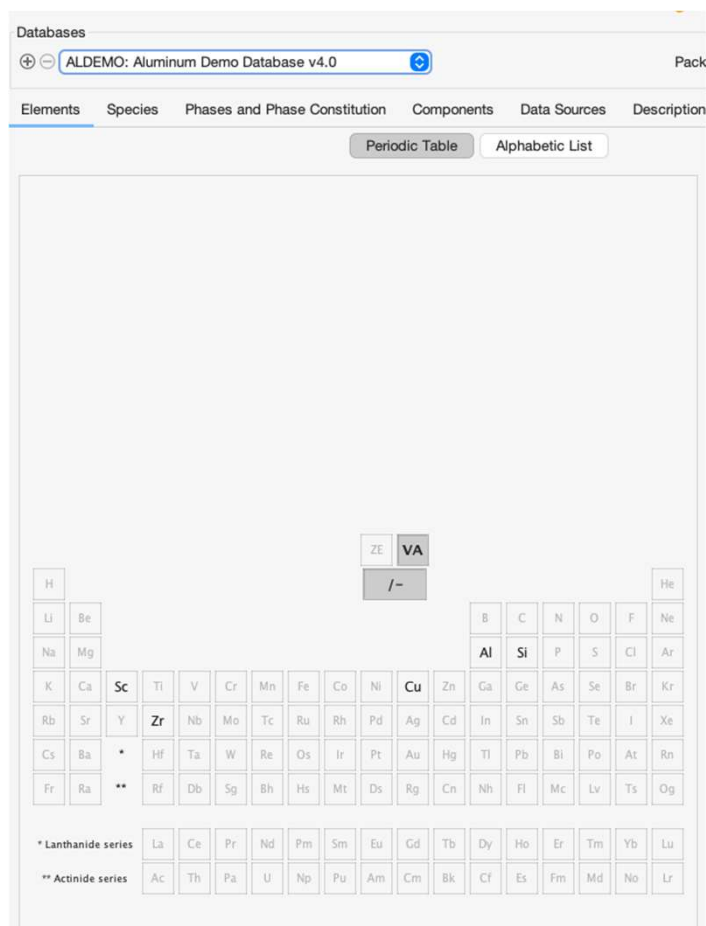
- Point calculations return the thermodynamic properties at a single condition
 - Stable phases
 - Enthalpy, Gibbs, etc

Project panel



- The project panel is the “map” for your calculation
- Calculations in Thermo-Calc in general follow the following:
 - Define system (System Definer)
 - Define conditions (Equilibrium Calculator)
 - Step/Map
 - Set axis/plot (Table Renderer)

System Definer



- Use the drop down “database” menu and select ALDEMO
- Select AL from the periodic table

Equilibrium Calculator

- Select equilibrium calculator from the project panel
- Fill it in like below

 **Equilibrium Calculator 1**

Conditions Functions Options

Composition unit Mass percent 

Condition Definitions

Temperature	Kelvin 	298.15
Pressure	Pascal 	101325
System size	Mole 	1.0
Composition	Al	100.0

Calculation Type

☒ Single equilibrium
 ☐ One Axis
 ☐ Grid
 ☐ Phase diagram

Results:

Visualizations

Table Renderer 1

System				
Moles	1.00000			
Mass	26.98200	[g]		
Temperature	298.15000	[K]		
Total Gibbs Energy	-8437.63272	[J]		
Enthalpy	0.01103	[J]		
Volume	9.99597E-6	[m ³]		

Component	Mole Fraction	Mass Fraction	Activity	Potential
Al	1.00000	1.00000	0.03325	-8437.63272

Stable Phases

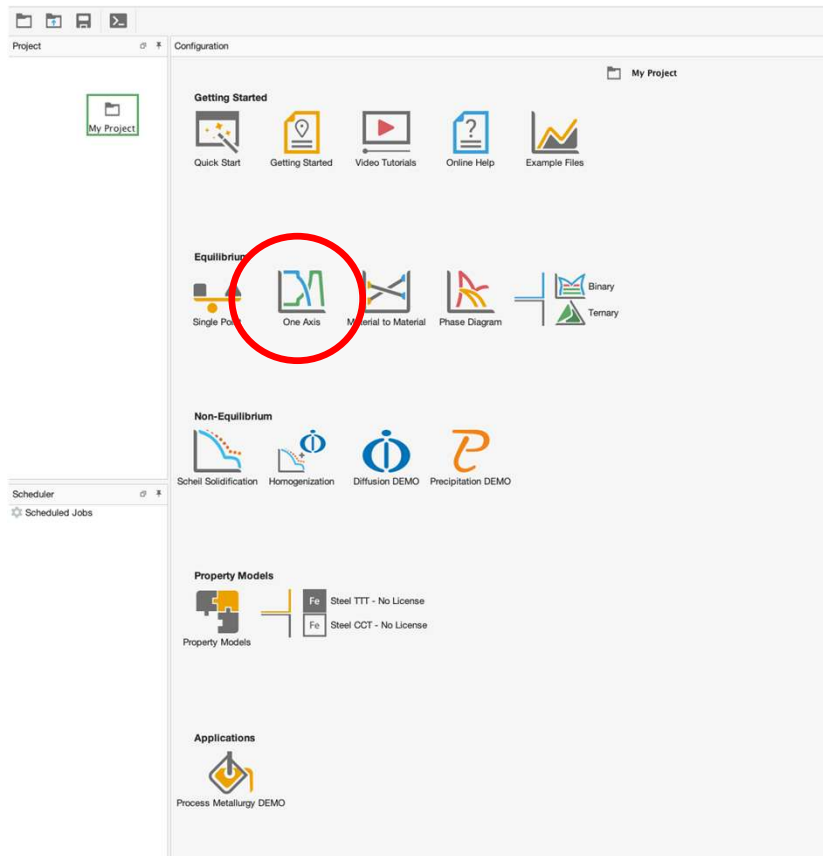
	Moles	Mass	Volume Fraction	
FCC_A1#1	1.00000	26.98200	1.00000	Composition

Composition

Component	Mole Fraction	Mass Fraction
Al	1.00000	1.00000

- Select Perform to get results
- 
- To the right are the results of our calculations in Table Renderer
 - We can see common thermodynamic properties of our system

Property Diagrams:



- Thermocalc 2023 changed name “property diagrams” to “One-Axis”
- Property diagrams plot a thermodynamic property vs a shift in conditions
- Shifts in condition include:
 - Temperature
 - Pressure
 - Composition
- Thermodynamic properties include
 - Gibbs
 - Enthalpy
 - Entropy
 - Site fractions
 - User defined functions

System Definer/Equilibrium calculator:

- Once again select Al from ALDEMO database
- Fill in the equilibrium calculator as such

Conditions
Functions
Options

Composition unit
Mass percent

Condition Definitions

Temperature
Kelvin
298.15

Pressure
Pascal
101325

System size
Mole
1.0

Composition
Al
100.0

Calculation Type

☐ Single equilibrium
☒ One Axis
☐ Grid
☐ Phase diagram

Axis Definitions

Quantity	Min	Max	Step division	Type	Step Method
Temperature	298.15	1298.15	50.0	Linear - min no. of steps	Normal

Plot Renderer:

- Change your plotting settings:
- Select enthalpy
 - Can select other properties as well
- Make sure units are “per mole”

Plot Renderer 1

Equilibrium Calculator 1

Legend option: Axis quantity Legend style: None Flexible Mode

X-axis

Axis variable: Temperature Kelvin

Axis type: Linear

Limits: 500.0 to 3000.0 step 250.0 Automatic scaling

Y-axis

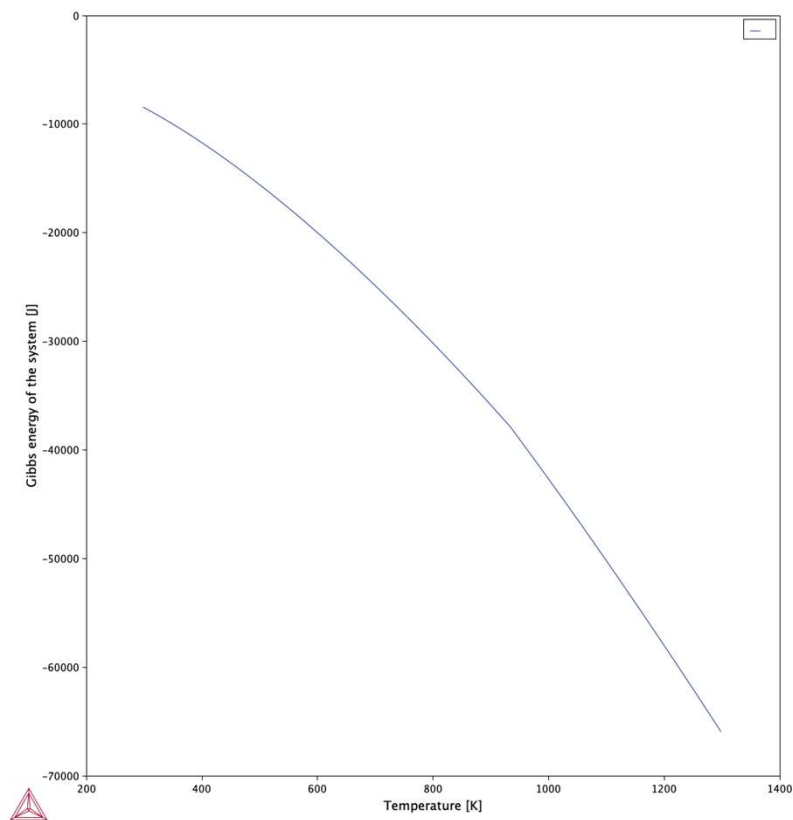
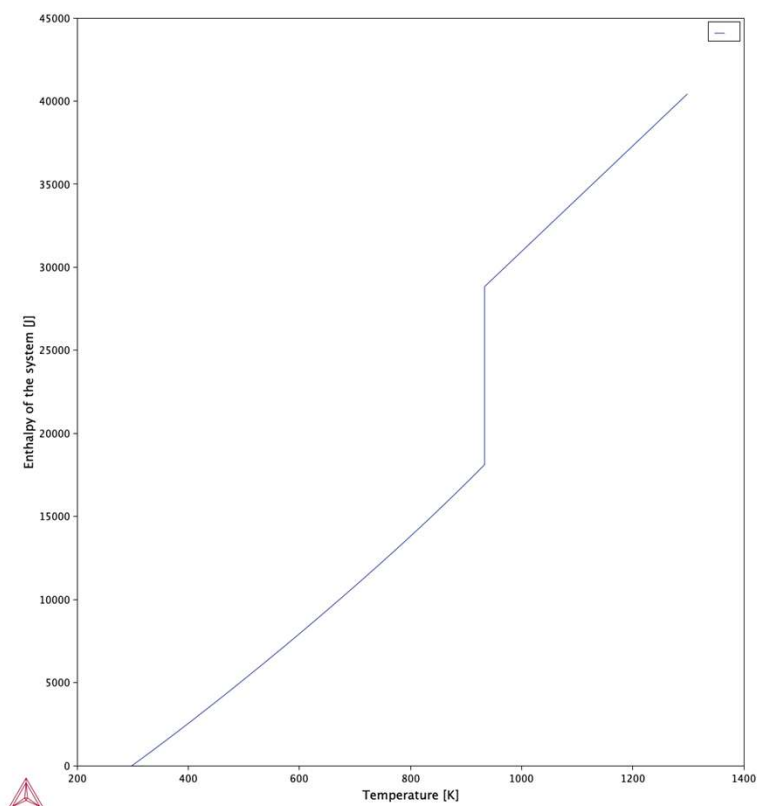
Axis variable: Enthalpy No normalization SER System Joule

Axis type: Linear

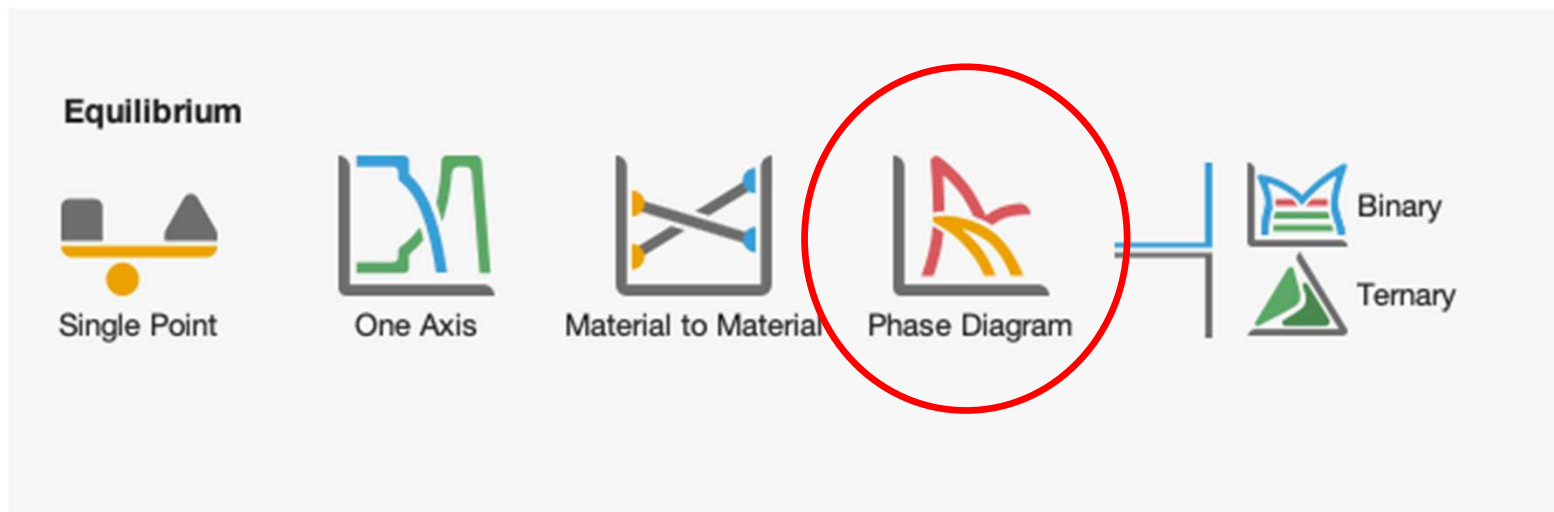
Limits: 0.0 to 1.0 step 0.1 Automatic scaling

+ - ✓

Resulting Plots:



Phase Diagrams



P-T diagram for Ni

1. System definer:
 - Set Ni from FEDEMO
2. Equilibrium calculator:

Equilibrium Calculator 2

Conditions Functions Options

Composition unit: Mass percent Switch to Advanced Mode

Condition Definitions

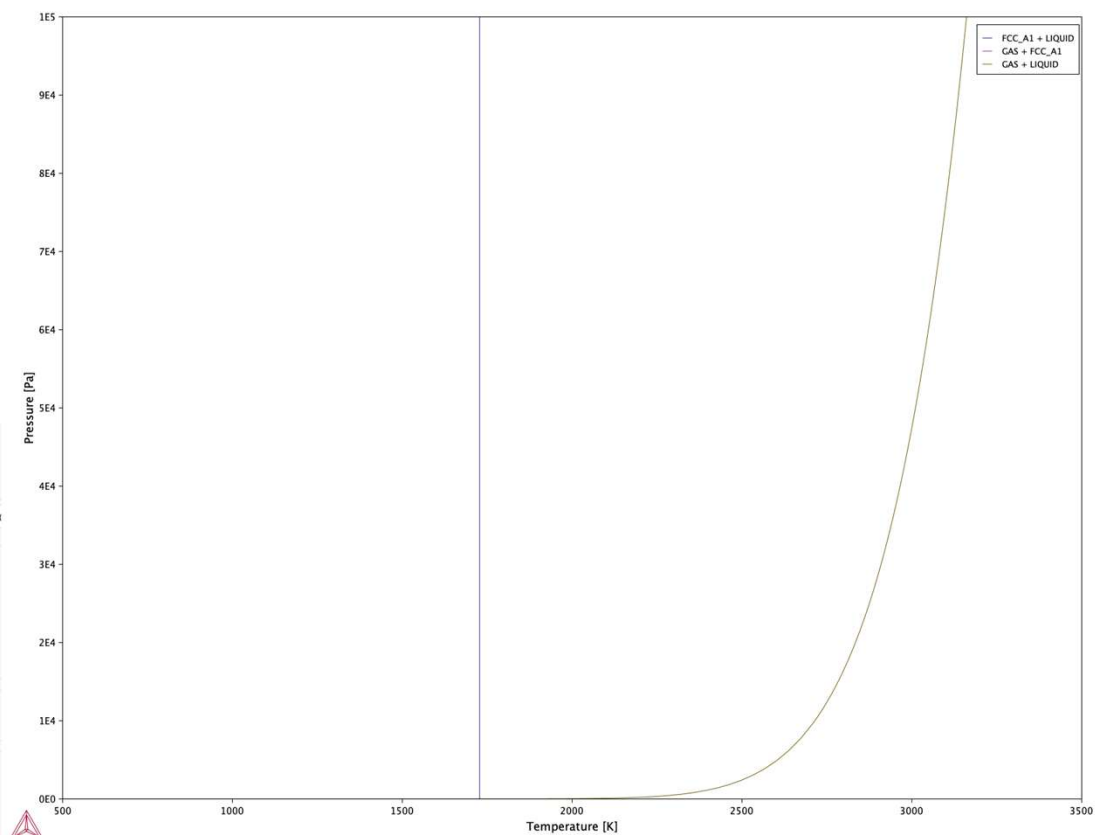
Temperature	Kelvin	298.15
Pressure	Pascal	101325
System size	Mole	1.0
Composition	Cu	100.0

Calculation Type

☐ Single equilibrium
 ☐ One Axis
 ☐ Grid
 ☒ Phase diagram

Axis Definitions

Quantity	Min	Max	Step division	Type
Temperature	200	2500	100	Linear - min no. of steps
Pressure	0.0	100000.0	50.0	Logarithmic - scale factor



Binary phase diagram

1. System Definer:
 - Fe-Cr from FEDEMO
2. Equilibrium Calculator:

Conditions

Functions

Options

Composition unit
 Mass percent

Condition Definitions

Temperature

Kelvin

298.15

Pressure

Pascal

101325

System size

Mole

1.0

Composition

Fe

99.99

Composition

Cr

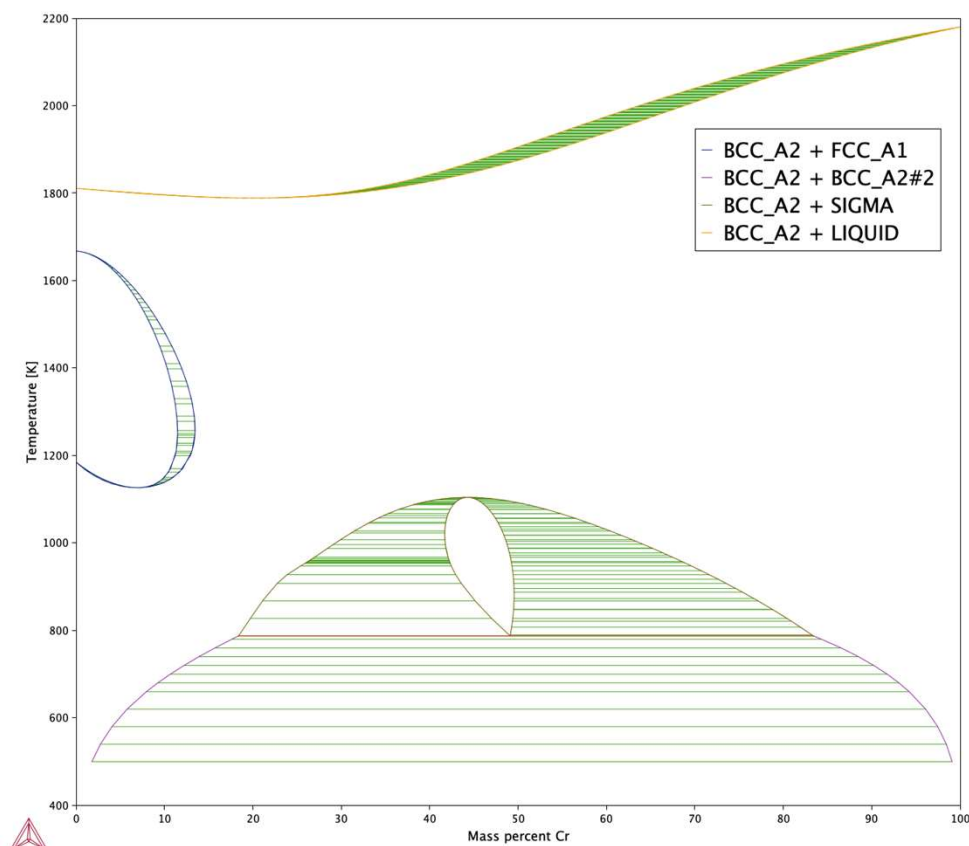
0.01

Calculation Type

☐ Single equilibrium
 ☐ One Axis
 ☐ Grid
 ☒ Phase diagram

Axis Definitions

Quantity	Min	Max	Step division	Type
Mass percent Cr	0.0	100.0	100	Linear - min no. of steps
Temperature	500.0	2500	50.0	Linear - min no. of steps



Ternary Phase Diagrams

Equilibrium



Single Point



One Axis



Material to Material



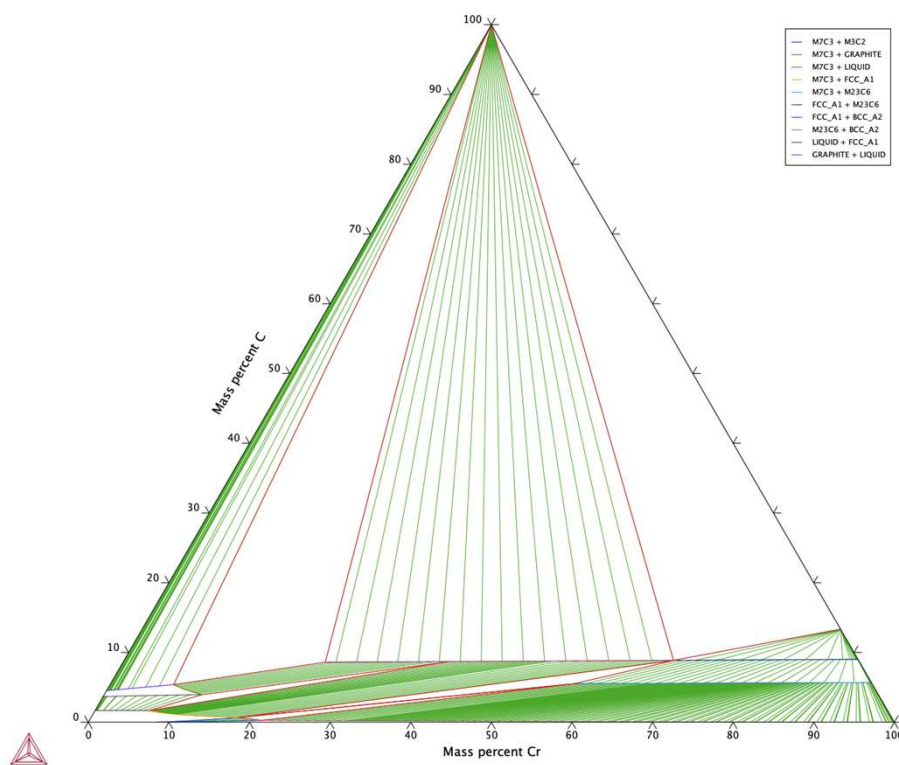
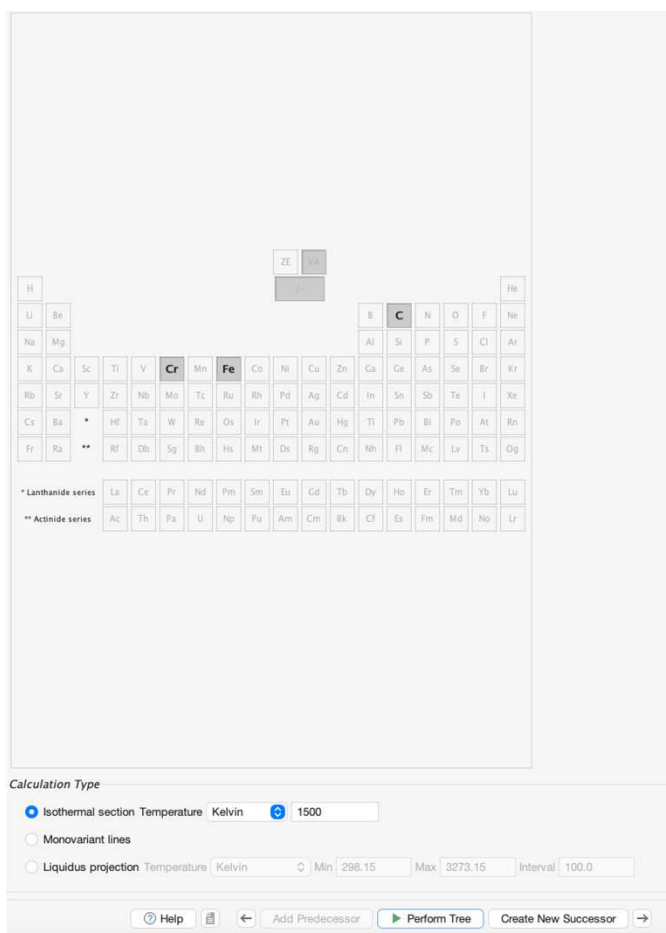
Phase Diagram



Binary

Ternary

Isothermal section: C-Cr-Fe at 1500K



Liquidus projection: Cr-Fe-Ni

Periodic table showing elements Cr, Fe, and Ni highlighted.

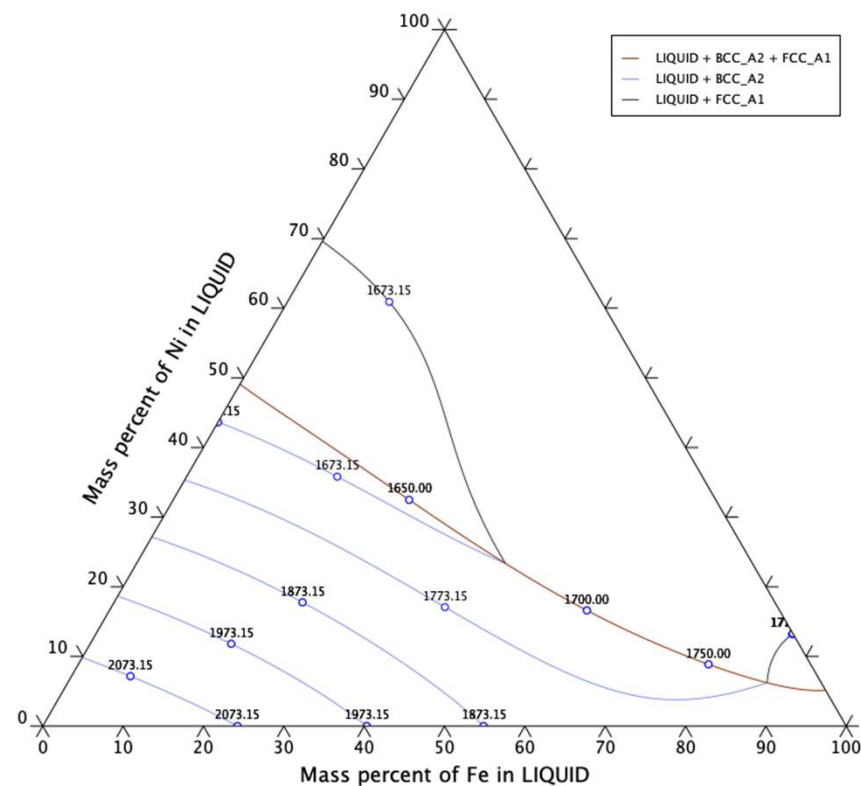
Calculation Type

☐ Isothermal section Temperature Kelvin 1500

☐ Monovariant lines

☒ Liquidus projection Temperature Kelvin Min 900 Max 3000 Interval

Help Add Predecessor Perform Tree Create New Successor

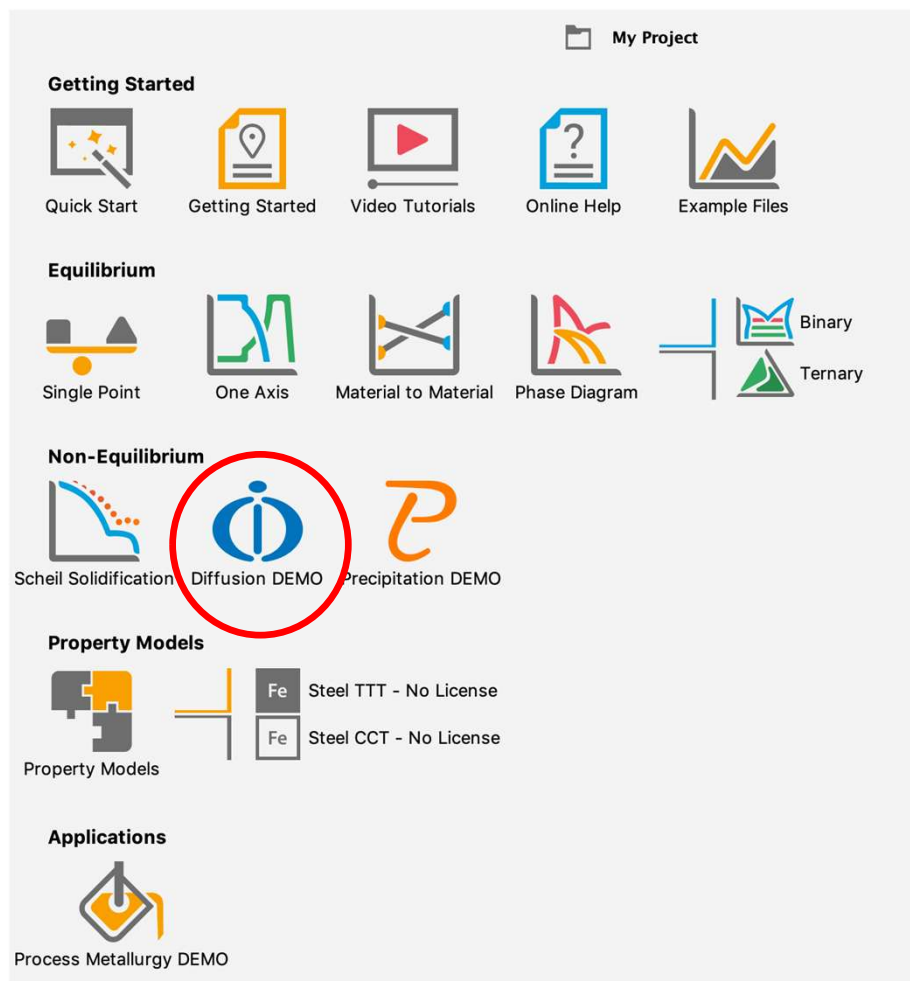




DICTRA

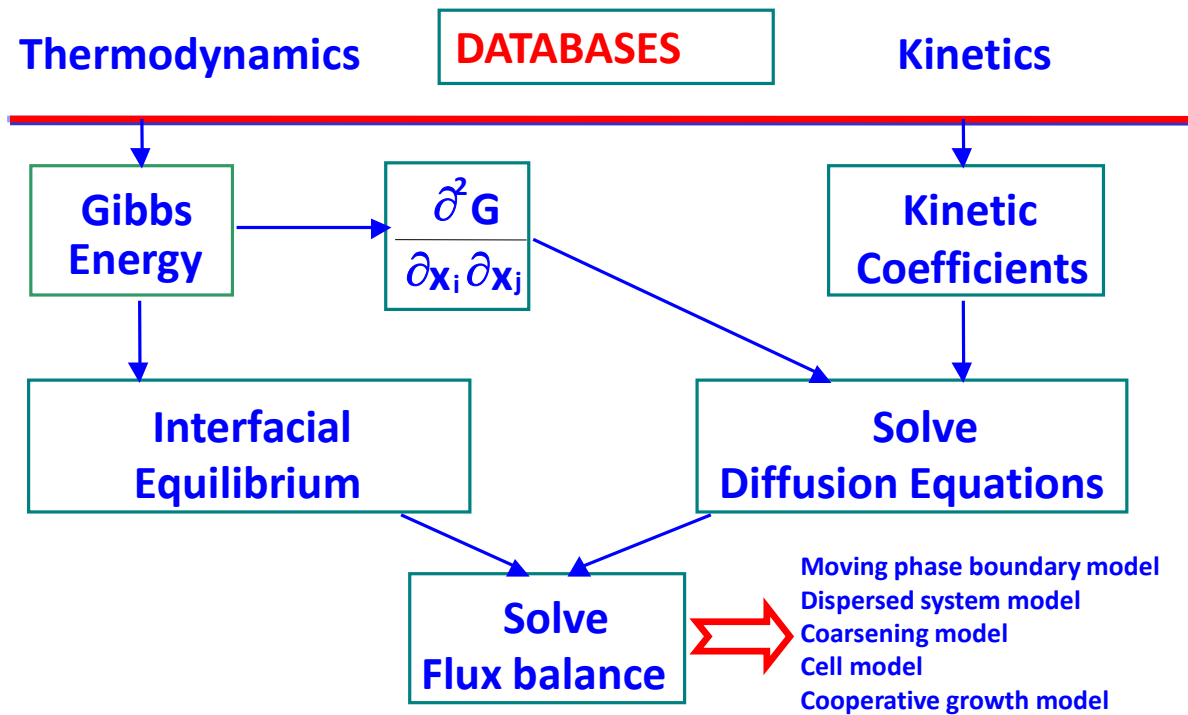
Diffusion-controlled Transformations Software

Thermo-Calc graphical mode templates



- Software for diffusion simulations
- One dimensional: planar, cylindrical, spherical
- Quantitative description of multicomponent diffusion
- Simulations based on databases for thermodynamics and diffusion
- Graphical mode for DICTRA was introduced in Thermo-Calc 2019a

Numerical simulations based on thermodynamic and mobility databases



General steps to set up DICTRA simulations

- Select geometry
 - One dimensional code: composition varies along one special coordinate
 - Planar: an infinitely wide plate of a certain thickness
 - Spherical: a sphere with a certain radius
 - Cylindrical: an infinitely long cylinder of a certain radius

Composition unit:

Length unit:

Geometry: 

Geometry: 

Geometry: 

General steps to set up DICTRA simulations

- Select geometry
- Define regions
 - Phase
 - Grid type
 - Composition profile

Region1

Region1

Left Interface Boundary

▼ Show left interface boundary

⊕ ⊖ Region

Name: Region1

Width: 1 m 50 points and type Linear

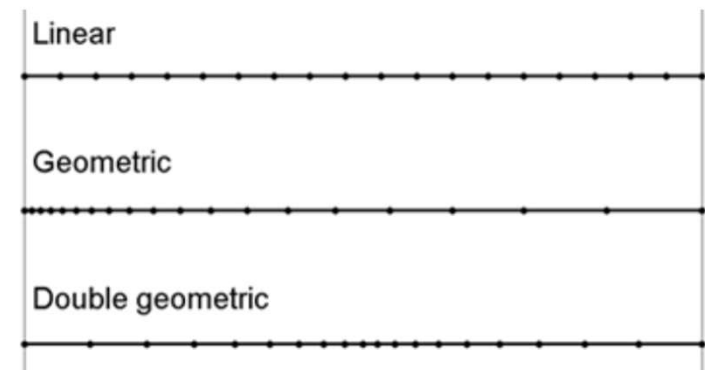
⊕ ⊖ Phase: None

Composition profiles: Edit

Linear

Geometric

Double geometric



General steps to set up DICTRA simulations

- Select geometry
- Define regions
- Select boundary conditions
- Define thermal profile

The screenshot displays the DICTRA simulation setup interface. It features two main sections for boundary conditions and a section for the thermal profile.

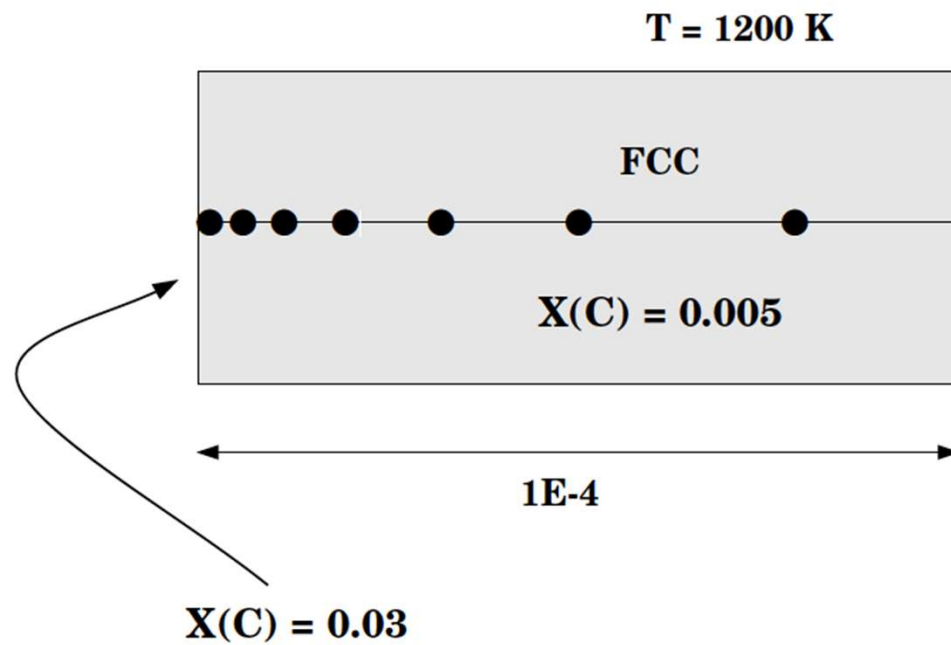
Left Boundary Condition: A dropdown menu is set to "Closed system".

Right Boundary Condition: A dropdown menu is open, showing three options: "Closed system" (highlighted in blue), "Mixed zero flux and activity", and "Composition".

Thermal Profile: Two radio buttons are present: "Isothermal" (selected) and "Non-isothermal". Below these, there are two rows of input fields:

- Temperature:** The input field contains "0.0" and the unit dropdown is set to "Kelvin".
- Simulation time:** The input field contains "0.0" and the unit dropdown is set to "Seconds".

One phase problem: Carburization of a binary Fe-C alloy



One phase problem: Carburization of a binary Fe-C alloy

Databases

⊕ ⊖ FEDEMO: Iron Demo Database v4.0 Package: Demo: Steels and Fe-alloys (FEDEMO, MFEDEMO)

⊕ ⊖ MFEDEMO: Fe-Alloys Mobility demo database v2.0

Elements Species Phases and Phase Constitution Components Data Sources Description

Periodic Table Alphabetic List

Material
Material name:

Amount Mass percent

Fe 99.99

C 0.01

Load Material
Save Material As...

- Select database and elements:
Database: FEDEMO, MFEDEMO
Elements: Fe, C

One phase problem: Carburization of a binary Fe-C alloy

Conditions Options

Composition unit: Mass percent

Length unit: Meter

Geometry: Planar

Region1

Region1

Left Interface Boundary

Show left interface boundary

Region

Name: Region1

Width: 1e-4 m 50 points and type Linear

Phase: FCC_A1

Composition profiles: Given by function Table input

Dependent component: Fe

Component C: Linear from 0.005 to 0.005

Right Interface Boundary

Show right interface boundary

Left Boundary Condition

Composition Fe 99.97 C 0.03

Right Boundary Condition

Closed system

Thermal Profile

Isothermal Non-isothermal

Temperature: 1200 Kelvin

Simulation time: 100 Seconds

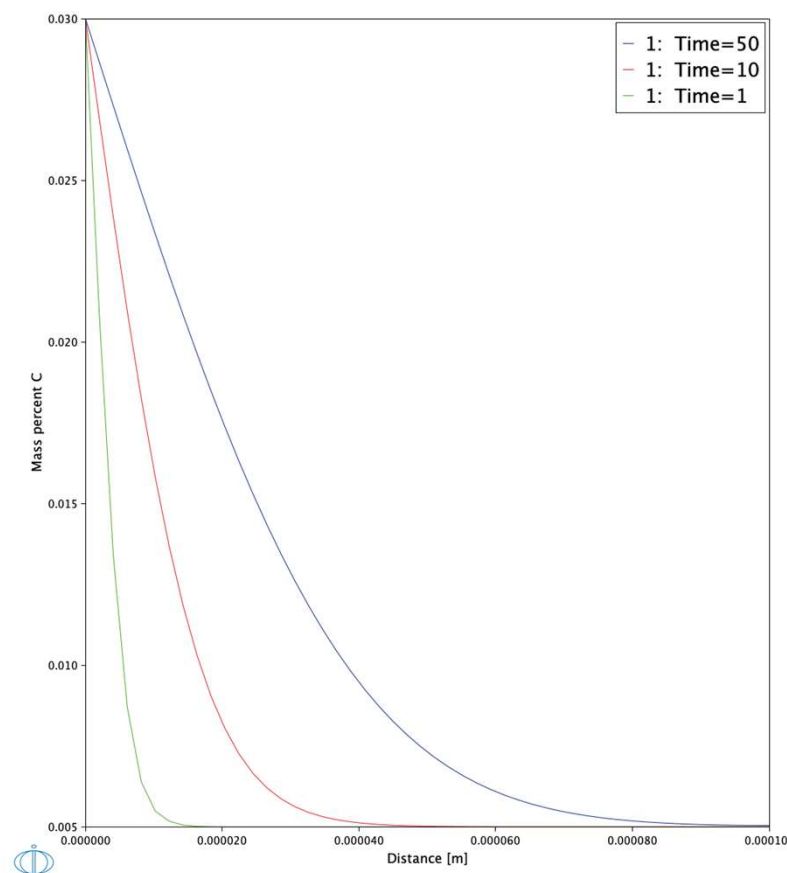
- Select geometry: Planar
- Define regions:
 - Width: 1e-4
 - Phase: FCC_A1
- Select boundary conditions
 - $X(C)=0.03$
- Define thermal profile
 - $T=1200K$
 - Time: 100s



One phase problem: Carburization of a binary Fe-C alloy

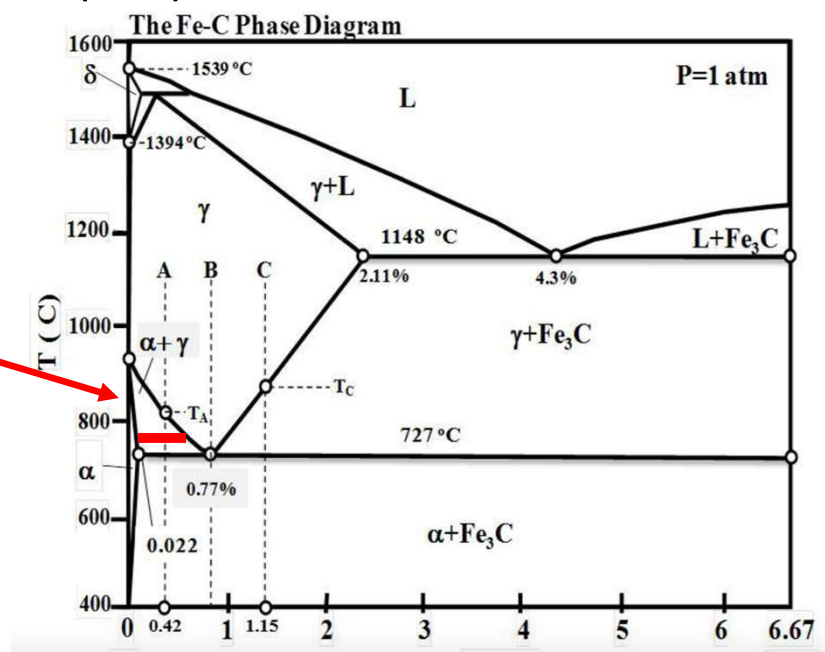
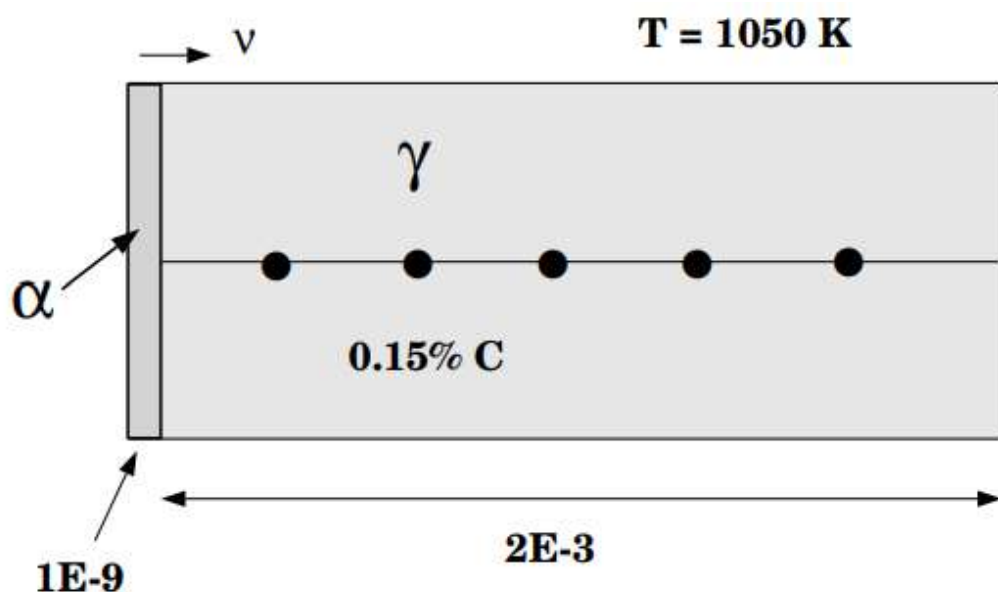


- Plot renderer
 - X: Distance
 - Y: composition
 - Time: 1s, 10s, 50s

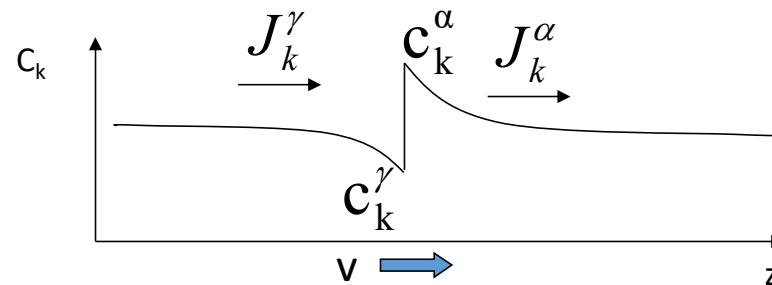


Fe-C isothermal moving boundary simulation

- Simulate the growth of ferrite (BCC) to austenite (FCC)



One-dimensional moving phase boundary



Unknowns: Tie-line, specified by $n-2$ activities a_i or chemical potentials μ_i

Velocity of phase boundary, v

Equations: $n-1$ flux-balance equations, $v(c_k^\alpha - c_k^\gamma) = J_k^\alpha - J_k^\gamma$

Solved as: $v(c_k^\alpha - c_k^\gamma) - (J_k^\alpha - J_k^\gamma) = 0$

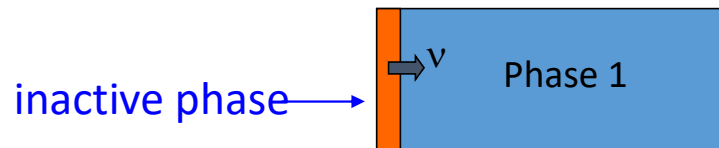
Set up for moving phase boundary

Moving phase boundaries simulations may be setup in DICTRA in two different ways:

- Introducing two or more adjacent regions containing different phases



- Entering an inactive phase (formed when becoming thermodynamically stable)



Fe-C isothermal moving boundary simulation

System with Fe C

Databases

⊕ ⊖ FEDEMO: Iron Demo Database v4.0 Package: Demo: Steels and Fe-alloys (FEDEMO, MFEDEMO) ⊖

⊕ ⊖ MFEDEMO: Fe-Alloys Mobility demo database v2.0

Elements Species Phases and Phase Constitution Components Data Sources Description

Periodic Table Alphabetic List



Material

Material name:

Amount Mass percent

Fe 99.99

C 0.01

Load material...

Save material as...

- Select package and elements
- The composition is not important to enter here, since the overall composition will be determined by setup of the regions

Fe-C isothermal moving boundary simulation

ConditionsOptions

Composition unit: Mass percent
Length unit: Meter
Geometry: Planar

Ferrite

Left Interface Boundary

Show left interface boundary

Region

Name: Ferrite
Width: 0 m 50 points and type Linear
Phase: None
Composition profiles: Edit

Right Interface Boundary

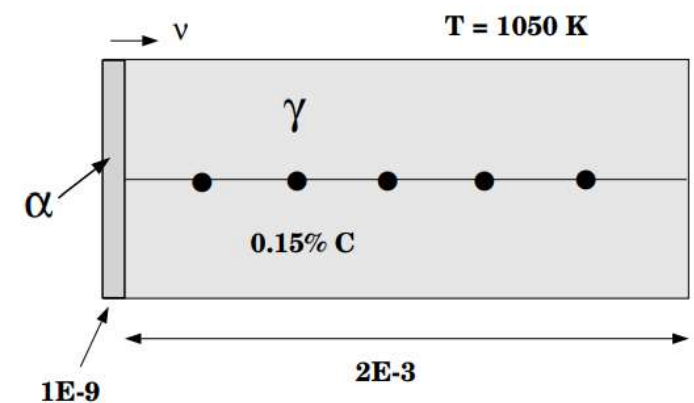
Show right interface boundary

Left Boundary Condition

Right Boundary Condition

Closed system
Closed system

- The default settings for the diffusion simulator are shown here



Fe-C isothermal moving boundary simulation

Conditions Options

Composition unit: Mass percent

Length unit: Meter

Geometry:  Planar

Ferrite

Left Interface Boundary

▼ Show left interface boundary

Region

Name: Ferrite

Width: 0 m 50 points and type Linear

Phase: None

Composition profiles: Edit

Right Interface Boundary

▼ Show right interface boundary

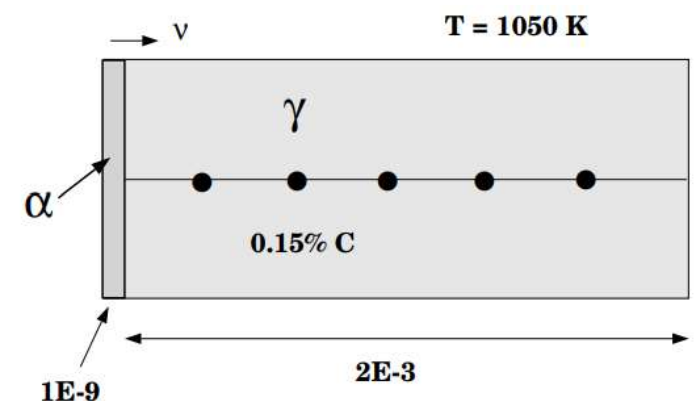
Left Boundary Condition

Closed system

Right Boundary Condition

Closed system

- Keep the Planar geometry



Fe-C isothermal moving boundary simulation

Conditions Options

Composition unit: Mass percent

Length unit: Meter

Geometry:  Planar

Ferrite

Ferrite

Left Interface Boundary

Show left interface boundary

Region

Name: Ferrite

Width: 1e-9 m 50 points and type Linear

Phase: BCC_A2

Composition profiles: Edit

Right Interface Boundary

Show right interface boundary

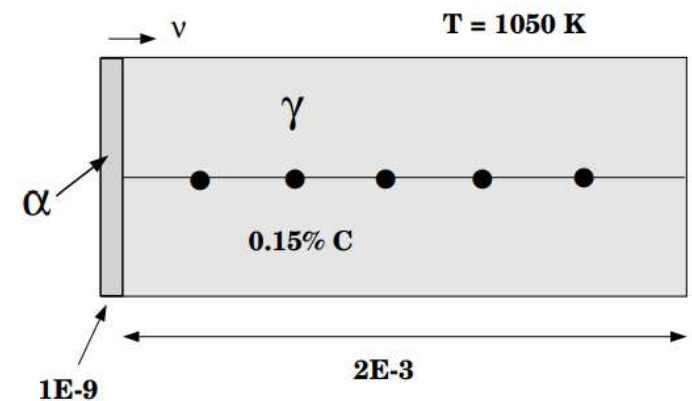
Left Boundary Condition

Closed system

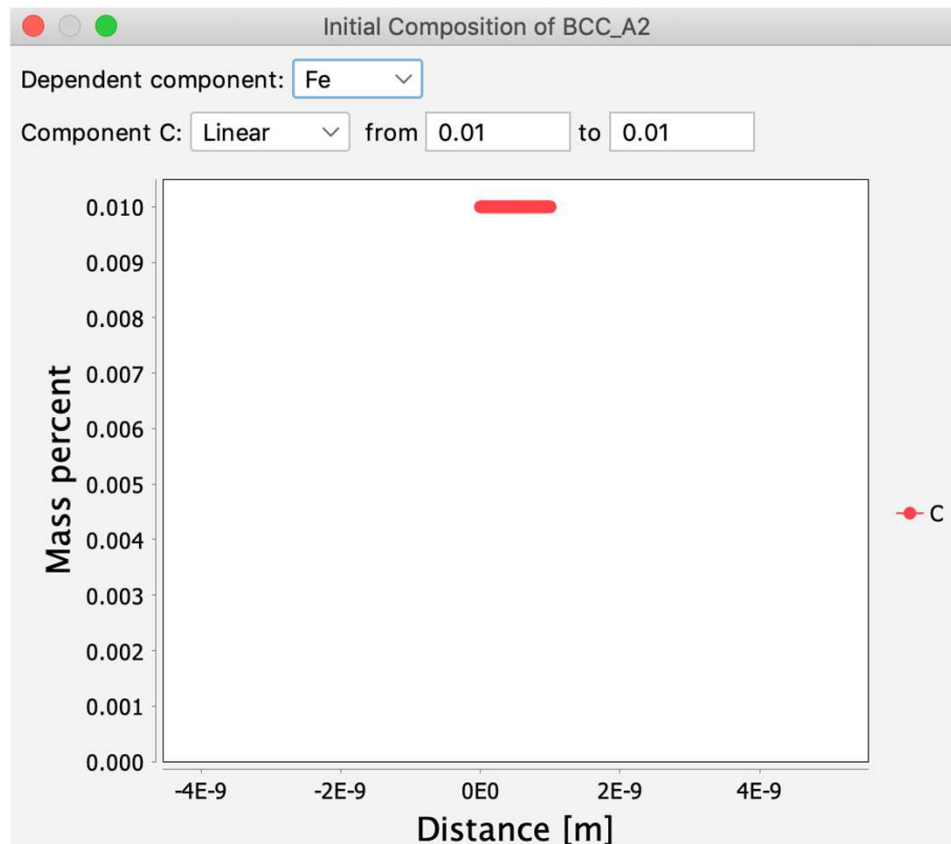
Right Boundary Condition

Closed system

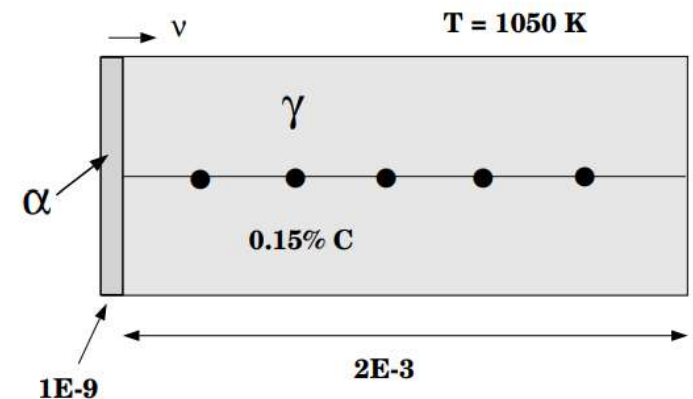
- Setup regions, for each:
 - Set the name, width and phase
 - “Edit” composition profile to set composition
- Geometry picture shown



Fe-C isothermal moving boundary simulation



- Edit composition window
- Can do complicated compositions, here we choose a constant, 0.01% C in Ferrite



Fe-C isothermal moving boundary simulation

Conditions Options

Composition unit: Mass percent

Length unit: Meter

Geometry:  Planar

.. Austenite

Ferrite Austenite

Left Interface Boundary

▼ Show left interface boundary

⊕ ⊖ Region

Name: Austenite

Width: 2e-3 m 50 points and type Linear

⊕ ⊖ Phase: FCC_A1

Composition profiles: Edit

Right Interface Boundary

▼ Show right interface boundary

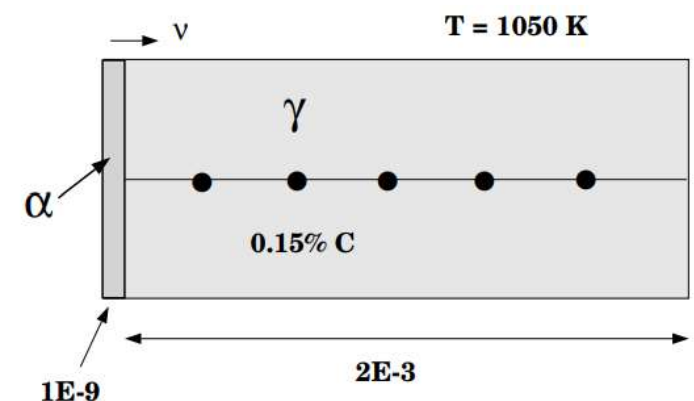
Left Boundary Condition

Closed system

Right Boundary Condition

Closed system

- Add additional regions with “+” button
- A new tab is created for the new region
- Setup Austenite the same way
- Use 0.15% C composition



Fe-C isothermal moving boundary simulation

ConditionsOptions

FerriteAustenite

Left Interface Boundary

Show left interface boundary

Region

Name: Austenite

Width: 2e-3 m 50 points and type Linear

Phase: FCC_A1

Composition profiles: Edit

Right Interface Boundary

Show right interface boundary

Left Boundary Condition

Closed system

Right Boundary Condition

Closed system

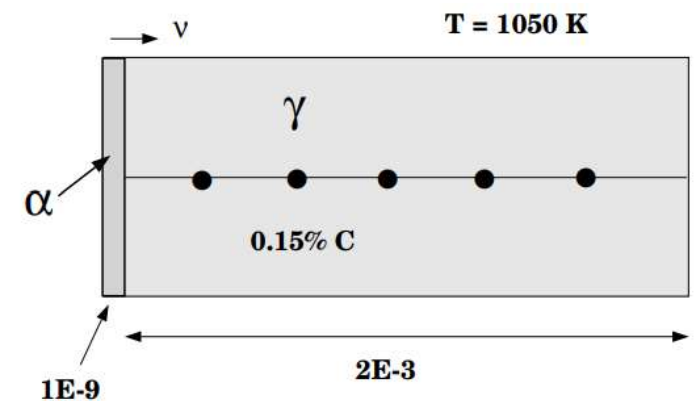
Thermal Profile

☒ Isothermal
 ☐ Non-isothermal

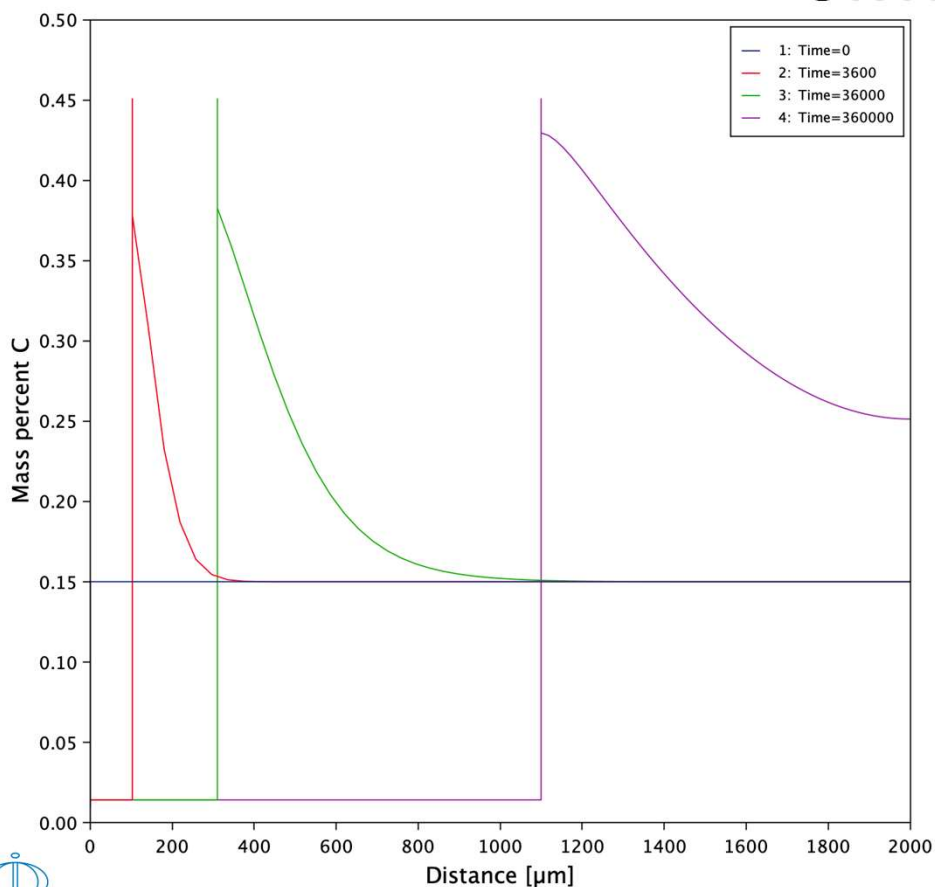
Temperature: 1050.0 Kelvin

Simulation time: 100 Hours

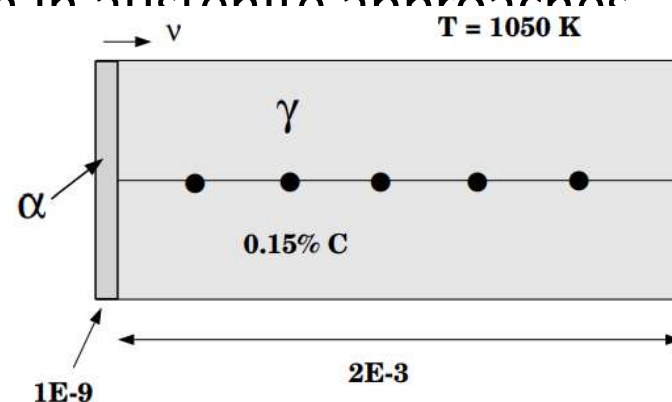
- Scroll down to see thermal profile
- Isothermal at 1050 K
- Perform the simulation



Fe-C isothermal moving boundary simulation



- Results for 0 h, 1 h, 10 h, 100 h
- Ferrite grows (interface moves right)
- Composition in ferrite stays low (tie-line)
- Interface line corresponds to local equilibrium tie-line compositions
- Composition in austenite approaches the tie-line



Fe-C isothermal moving boundary simulation

Diffusion two regions Ferrite and Austenite

Simple mode

X-axis

Axis variable: Distance Micrometer Region All regions

Axis type: Linear

Limits: 0.0 to 2000.0 step 200.0 ☐ Automatic scaling

Y-axis

Axis variable: Composition C Mass percent

Axis type: Linear

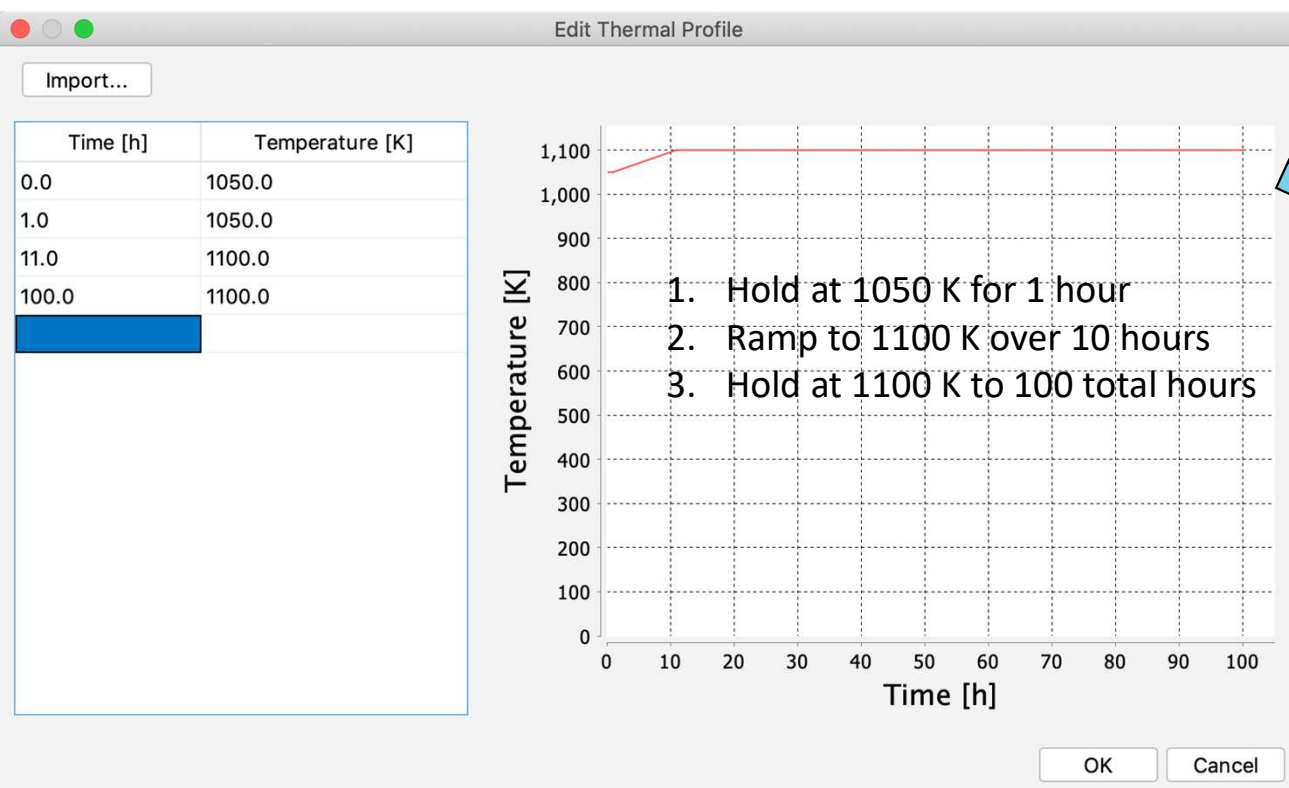
Limits: 0.0 to 100.0 step 10.0 ☒ Automatic scaling

Time Hours Value(s) 0.0 1.0 10.0 100.0

+ - ☒

- To plot, we'll plot the composition profile at several times
- Plot at 0 hours (initial), 1, 10, 100

Fe-C non-isothermal sim



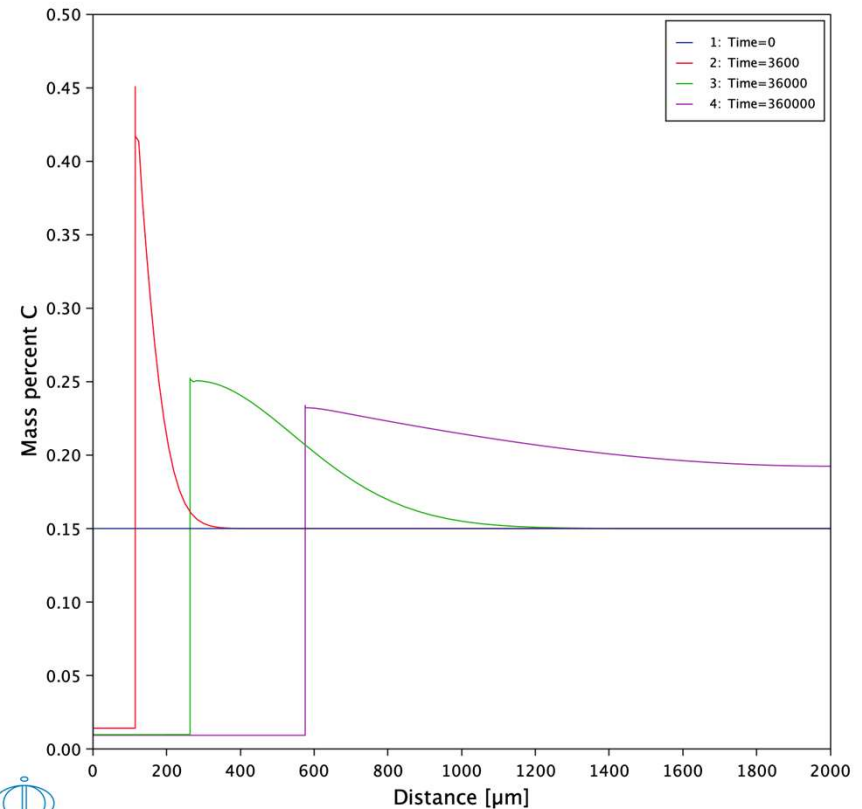
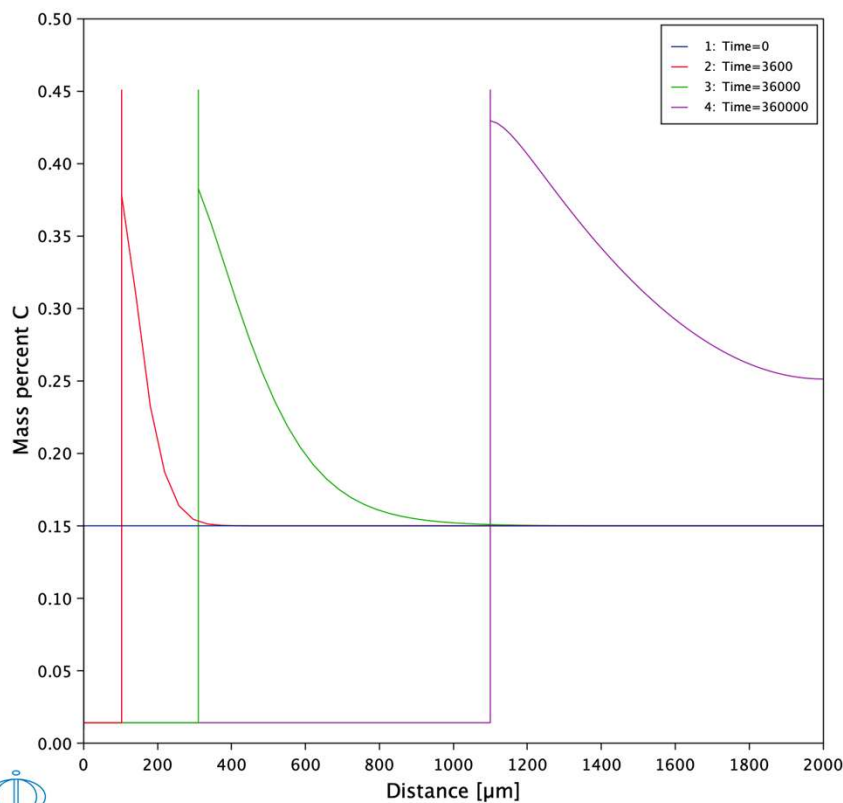
The 'Thermal Profile' dialog box shows the following settings:

- ☐ Isothermal
- ☒ Non-isothermal
- Temperature unit: Kelvin
- Time unit: Hours
- Edit thermal profile** (button)
- Simulation time: 100.0

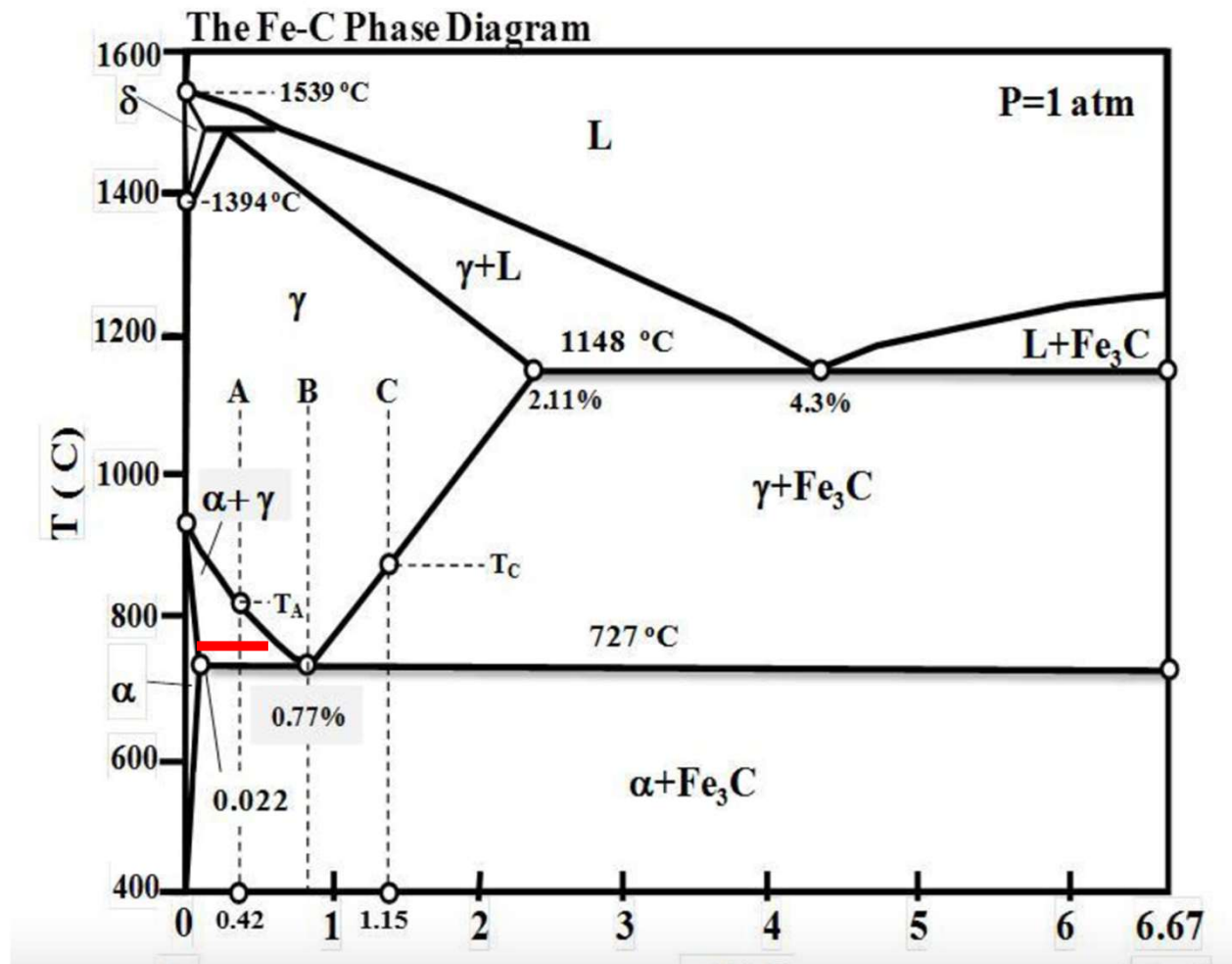
- Non-isothermal example
- Choose non-isothermal and then “Edit thermal profile” button
- Each row is a point on the curve, i.e., the temperature at a time
- The curve ramps between temperatures
 - repeated T = isothermal

Comparison of thermal diffusion profiles

- The first hour is (about) the same (the same temperature profile)
- Increasing the temperature caused use to reach equilibrium faster
- But the composition of C in both phases decreased



Comparison of thermal diffusion profiles

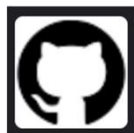




GitHub and Codespaces

- GitHub:

- A **developer platform** that to create, store, manage and share their code. It uses Git software, providing the distributed version control, enabling multiple individuals to work on projects simultaneously.



<https://www.github.com/>

- Codespaces

- **Cloud-Based Development Environments:** A feature within GitHub that provides fully configurable, cloud-hosted development environments, allowing developers to code, build, test, and debug applications directly from their browser or through an integrated development environment (IDE) like Visual Studio Code.
- **Pre-configured for Collaboration:** Pre-configured for GitHub's collaboration features, enabling real-time collaboration among team members and providing a consistent environment for all contributors.

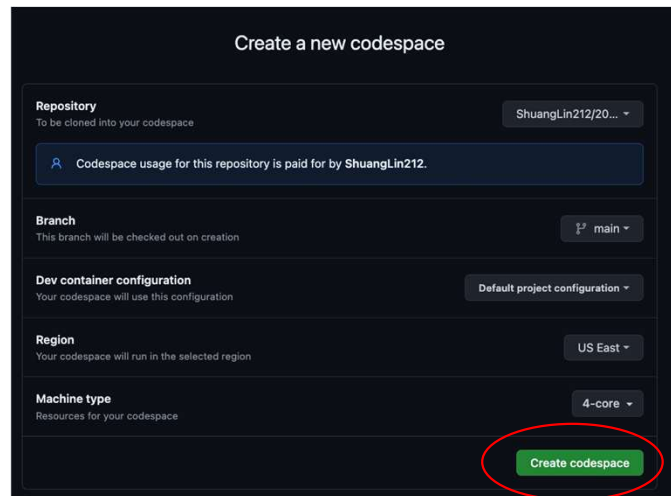
Codes on github

- Github website:

https://github.com/ShuangLin212/2024_MatSE410_lectures_computational_tools

- Codespaces link:

https://codespaces.new/ShuangLin212/2024_MatSE410_lectures_computational_tools



The screenshot shows the 'Create a new codespace' interface. It includes several configuration options: Repository (ShuangLin212/2024...), Branch (main), Dev container configuration (Default project configuration), Region (US East), and Machine type (4-core). A green 'Create codespace' button is highlighted with a red circle at the bottom right.